

Rural Development Policies in Southeast Asia: Impacts on Household Livelihoods, Welfare and Rural Transformation

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- *To my beloved father* -

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ZUSAMMENFASSUNG

Strukturwandel in der Landwirtschaft ist ein wesentlicher Bestandteil der wirtschaftlichen Entwicklung von Entwicklungsländern. Dieser Prozess ist gekennzeichnet durch die Verlagerung von Arbeitskräften aus der Landwirtschaft in nichtlandwirtschaftliche Sektoren, was zu höheren Einkommen aus nichtlandwirtschaftlichen Quellen führt. Innerhalb des Agrarsektors kann dieser Strukturwandel den Einsatz von Mechanisierung verändern und das Haushaltseinkommen aus landwirtschaftlichen Quellen verbessern. In Entwicklungsländern spielt die Landwirtschaft eine bedeutende Rolle für den Lebensunterhalt ländlicher Haushalte, daher kann dieser Strukturwandel auch als landwirtschaftliche Transformation oder ländliche Transformation betrachtet werden. Die Untersuchung, wie ländliche Entwicklungspolitik, beispielsweise die Förderung von Investitionen in die ländliche Infrastruktur, den Strukturwandel begleiten können, ist für eine erfolgreiche Transformation der Landwirtschaft und der ländlichen Gebiete von entscheidender Bedeutung.

Vietnam ist ein gutes Beispiel für die Untersuchung ländlicher Entwicklungspolitik und des Strukturwandels. Erstens ist dieses Land für sein schnelles Wirtschaftswachstum bekannt, wobei aber ein großer Teil der Bevölkerung des Landes in ländlichen Gebieten lebt und auf die Landwirtschaft als Haupteinnahmequelle angewiesen ist. Zweitens hat das Land seit 2010 offiziell eine neue Politik der ländlichen Entwicklung (NRD) eingeführt, um wesentliche Probleme in ländlichen Gebieten zu lösen. Im Jahr 2022 verlängerte die vietnamesische Regierung dieses nationale Programm um einen weiteren Zeitraum bis 2025. Daher stellt sich die Frage, ob die bisherigen Investitionen für die Verbesserung des Einkommens und des Wohlstands ländlicher Haushalte wirksam sind. Drittens und letztlich ist die ländliche Bevölkerung in Vietnam aufgrund extremer Wetterereignisse und gesundheitlicher Schocks noch anfälliger. Diese Dissertation zielt daher darauf ab, empirische Beweise dafür zu liefern, wie die NRD-Politik dazu beitragen kann, das Einkommen ländlicher Haushalte zu erhöhen, ihren Wohlstand und ihre Resilienz gegen Schocks zu verbessern, Armut zu verringern und gleichzeitig den Strukturwandel im ländlichen Raum zu beschleunigen. Sollten Investitionen in die ländliche Infrastruktur zur Lösung wesentlicher Probleme in ländlichen Gebieten beitragen, ist es ein weiteres Ziel dieser Dissertation, Bereiche zu identifizieren, in denen ländliche Investitionen und politische Maßnahmen in Zukunft ansetzen sollten.

Die in allen Artikeln dieser Dissertation verwendeten Hauptdaten stammen vom Thailand - Vietnam Socio-Economic Panel (TVSEP). Der erste Artikel dieser Dissertation beantwortet die Frage, wie Verbesserungen in der ländlichen Infrastruktur und des

Einkommens zur Verbesserung der Resilienz ländlicher Haushalte gegenüber Schocks beitragen können. Der zweite Artikel untersucht, ob Beschäftigungsmöglichkeiten außerhalb der Landwirtschaft die Ausbeutung natürlicher Ressourcen und die Armut in ländlichen Gebieten verringern können. Die Ergebnisse zeigen, dass die Entwicklung der ländlichen Infrastruktur im Rahmen des NRD-Programms dazu beitragen kann, die Absorptions- und Anpassungsfähigkeit ländlicher Haushalte gegenüber Schocks zu erhöhen und die Armut in absoluter, relativer und mehrdimensionaler Hinsicht zu verringern. Eine bessere ländliche Infrastruktur trägt auch dazu bei, bessere Beschäftigungsmöglichkeiten außerhalb der Landwirtschaft zu schaffen, die wiederum zum Rückgang der Ausbeutung natürlicher Ressourcen und einer Verringerung der Armut führt. Der dritte und vierte Artikel konzentrieren sich auf mögliche politische Maßnahmen, die sowohl das Einkommen ländlicher Haushalte erhöhen als auch den Strukturwandel in ländlichen Gebieten anregen können. Die Ergebnisse zeigen, dass Maßnahmen zur Erleichterung der Landkonsolidierung und Förderung der Kommerzialisierung der Landwirtschaft den Strukturwandel anregen und Armut und Ungleichheit verringern können. Der fünfte und sechste Artikel weisen darauf hin, dass ländliche Investitionen mehr Wert auf die Gewährleistung einer ganzjährigen Bewässerung und die Verbesserung der Infrastruktur für Informations- und Kommunikationstechnologie (IKT) (d. h. die Internetkonnektivität) legen sollten. Diese Maßnahmen können dazu beitragen, das Wohlergehen der Menschen in ländlichen Gebieten und in benachteiligten Gruppen, wie etwa weiblichen Migranten, zu verbessern und die Armut zu verringern.

Neben den Beiträgen zur Literatur in Form empirischer Belege leistet diese Dissertation auch in zwei methodischen Aspekten einen Beitrag zur bestehenden Forschung. Erstens befasst sich diese Studie im zweiten Artikel mit der Gleichzeitigkeit von (i) nichtlandwirtschaftlichen Einkommen und Einkommen durch das Extrahieren natürlicher Ressourcen, im dritten Artikel mit (ii) landwirtschaftlicher Effizienz und Beteiligung an Flurbereinigungen und im vierten Artikel mit (iii) Kommerzialisierung und Mechanisierung in der Landwirtschaft. Zweitens befasst sich diese Studie in allen Artikeln mit Selektionsbias, nicht beobachtbaren Faktoren und Endogenitätsbedenken hinsichtlich der Variablen des Datensatzes mittels Auswahl von geeigneten ökonometrischen Methoden für Paneldaten.

Basierend auf den empirischen Belegen darüber, wie sich Haushaltskapital und dörfliche Umstände, insbesondere die Entwicklung der Infrastruktur, auf die Verbesserung der Resilienz ländlicher Haushalte, die Erhöhung des Einkommens, die Milderung der Auswirkungen von Schocks, die Verringerung der Armut und letztlich die Förderung des

Strukturwandels in Vietnam und Südostasien im Allgemeinen auswirken können, schlägt diese Studie mehrere politische Maßnahmen zur Stimulierung des Strukturwandels vor.

Stichworte: Strukturwandel; ländliche Lebensgrundlagen; Resilienz; Armutsbekämpfung; Wohlfahrt; Ungleichheit.

ABSTRACT

Structural change in agriculture is an essential part of economic development in developing countries. The process of structural change is characterized by the shift of labor from agriculture to non-agriculture sectors which leads to higher income from non-farm sources. Within the agricultural sector, structural change can influence the application of mechanization and improve income from farm sources of households who remain in the sector. In developing countries, agriculture plays a significant role in rural households' livelihoods, hence, structural change can also be considered as agricultural transformation or rural transformation. Studying how rural development policies, for instance promoting investments in rural infrastructure, can accompany the structural change is vital for a successful transformation of agriculture and rural areas.

Vietnam is a good example to study rural development policies and structural change. First, this country is famous for its rapid economic growth, but a large proportion of the country's population live in rural areas and rely on agriculture as the main income source. Second, the country has officially introduced a new rural development (NRD) policy since 2010 to solve critical issues in rural areas. In 2022, the government of Vietnam extended this national program for the next phase, up to 2025. Hence, the question arises whether the past investments are significant for improving rural households' income and welfare. Finally, the rural population in this country is even more vulnerable due to extreme weather events and health shocks. Therefore, this dissertation aims to provide empirical evidence on how this NRD policy can help increase rural households' income, improve their welfare and resilience against shocks, and reduce poverty, while accelerating structural transformation in rural areas. If investments in rural infrastructure can help solve critical problems in rural areas, this dissertation also aims to shed light on which areas of the rural investments and policy improvements should be implemented in the future.

The main data used in all papers of this dissertation come from the Thailand – Vietnam Socio-Economic Panel (TVSEP). The first paper of this dissertation answers how improvements in rural infrastructure and income contribute to the enhancement of rural households' resilience in the context of shocks. The second paper investigates whether non-farm employment opportunities can reduce the exploitation of natural extraction and poverty in rural areas. The results show that the development of rural infrastructure under the NRD program can help increase rural households' absorptive and adaptive capacities against shocks and reduce poverty in absolute, relative, and multidimensional terms. Better rural infrastructure

also helps provide better non-farm employment opportunities which contribute to the reduction of natural resource extraction and poverty. The third and fourth paper focus on potential policy improvements which can jointly increase rural households' income and stimulate structural change in rural areas. The results indicate that interventions in facilitating land consolidation and promoting agricultural commercialization can stimulate structural change and reduce poverty and inequality. The fifth and sixth paper point out that rural investments can place more emphasis on ensuring year-round irrigation and improving infrastructure for information and communication technology (ICT) (i.e., the internet connectivity). These improvements can help increase welfare and reduce poverty among people in rural areas and disadvantaged groups such as female migrants.

In addition to contributing to the literature in terms of empirical evidence, this dissertation also contributes to the literature in two methodological aspects. First, this study addresses the simultaneity of (i) non-farm income and income from natural extraction in the second paper, (ii) farming efficiency and participation in land consolidation in the third paper, and (iii) crop commercialization and mechanization in the fourth paper. Second, this study accounts for selection bias, unobservable factors, endogeneity concerns in all of papers by using panel data and appropriate econometric approaches.

Based on the empirical evidence on how household capital and village conditions, especially infrastructure development, can improve rural household resilience capacities, increase income, mitigate the effects of shocks, reduce poverty, and ultimately promote structural change in Vietnam and Southeast Asia in general, several policy implications are proposed to promote structural transformation.

Keywords: Structural change; rural livelihoods; resilience; poverty reduction; welfare; inequality.

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LIST OF ABBREVIATIONS

2SLS	Two Stages Least Squares
CF	Control function
DD	Difference-in-Differences
DFG	German Research Foundation
FE	Fixed-effects
GSO	General Statistics Office of Vietnam
Hetero-IV	Heteroscedasticity-based identification
ICT	Information and communication technology
IV	Instrumental variable
NRD	New Rural Development
PPP	Purchasing Power Parity
PSM	Propensity Score Matching
TRMM	Tropical Rainfall Measuring Mission
TVSEP	Thailand – Vietnam Socio - Economic Panel
UQR	Unconditional Quantile Regression
US\$	United States dollar
VND	Vietnam Dong

CHAPTER 1: INTRODUCTION

1.1 Motivation

Structural change in agriculture is an essential component of economic development in developing countries (Pingali and Abraham, 2022; Timmer, 1988). The process of this structural change is characterized by the shift of labor from agriculture to non-agricultural sectors, leading to higher incomes from non-farm sources (Johnston and Mellor, 1961; Lewis, 1954). Within the agricultural sector, structural change can influence the use of mechanization and improve the income from farm sources of households that remain in the sector (Amare et al., 2023). In developing countries, agriculture plays an important role in the livelihoods of rural households, hence, structural change can also be considered as agricultural transformation or rural transformation. Overall, structural change results in income growth and economic development (McMillan et al., 2014; Pingali et al., 2019; Timmer, 2017). Studying how rural development policies such as promoting investments in rural infrastructure, can accompany structural change is crucial for a successful transformation of agriculture and rural areas (Cervantes-Godoy, 2022).

Since *Doi Moi* in 1986, Vietnam has undergone a significant and positive transformation into an open, socialist-oriented market economy with some of the fastest economic growth in the world (Nguyen, 2012). However, the country has also been struggling to address critical issues in rural areas such as the overexploitation of natural resources, the widening income gap between rural and urban areas, and the higher poverty rate of the rural population. In 2010, the government of Vietnam officially launched a New Rural Development (NRD) program (i.e., a new rural development policy) to address the above issues and mobilized about 850 trillion Vietnam Dong (VND) (an equivalent to about US\$38 billion) for investments in rural development projects in all villages in the period from 2010 to 2015 (Do and Park, 2018). The central government of Vietnam has set two major targets for the NRD program, including (i) the improvement of rural infrastructure for production and livelihood purposes (such as roads, irrigation, electricity, and telecommunications) and (ii) the improvement of rural household income. To some extent, the NRD policy can be considered as a rural development policy aimed at improving the income of rural households and stimulating rural transformation in the country. However, empirical evidence on how investments in rural infrastructure and the

program itself can contribute to the process of rural transformation is scarce.

Vietnam is a good example to study rural development policies and rural transformation. First, this country is known for its rapid economic growth, but a large proportion of the country's population lives in rural areas and relies on agriculture as its main source of income (Nguyen et al., 2021). Second, the country officially introduced a new rural development policy in 2010 to solve critical issues in rural areas after a pilot phase between 2006 and 2009 (Do and Park, 2019). In 2022, the government of Vietnam extended this national program for the next period up to 2025. Hence, the question arises whether the past investments are significant for improving rural household income and welfare. Finally, the rural population of this country is even more vulnerable due to extreme weather events and health shocks (Nguyen et al., 2022b; Nguyen et al., 2023). Therefore, this dissertation aims to provide empirical evidence on how this NRD policy can help increase rural household income, improve their welfare and resilience against shocks, and reduce poverty, while accelerating structural transformation in rural areas. If investments in rural infrastructure can help solve critical problems in rural areas, this dissertation also aims to shed light on which areas of the rural investments and policy improvements should be implemented in the future.

1.2 Research objectives

Against this background, this dissertation aims to answer the following research questions:

1. How do improvements in rural infrastructure and income contribute to increasing the resilience of rural households in the context of shocks?
2. Can the creation of more non-farm employment opportunities reduce the exploitation of natural resources and poverty?
3. Is there potential for improvement in public policy to jointly increase income, stimulate rural transformation, and reduce poverty?
4. Is there potential for intervention to support structural change?
5. Is there potential scope for investment in rural infrastructure to improve household income and reduce poverty, especially in the context of shocks?
6. Will rural infrastructure development in information and communication technology (ICT) have an effect on the welfare of people in disadvantaged groups such as female migrants?

1.3 Methodologies

This dissertation employs a number of econometric models and methodologies for the empirical analyses that are explained as follows.

In the **first paper**, a household fixed-effects (FE) estimation model with instrumental variable (IV) is used to address the endogeneity concerns of the household head risk attitude in order to identify the factors affecting household resilience strategies, namely accumulating savings as absorptive capacity and diversifying income as adaptive capacity. Then, to examine the impacts of these household resilience strategies on shocks and poverty, we employ household fixed-effects models with IV to address the endogeneity issues of absorptive and adaptive capacity variables. We use the precipitation data from the Tropical Rainfall Measuring Mission (TRMM) to generate the IV for these estimations. As a robustness check, we apply an IV approach following the heteroscedasticity-based identification strategy suggested by Lewbel (2012) using the same IV as in the IV-FE estimations. The diagnostic tests and corresponding econometric models are conducted using STATA 15.

Regarding the **second paper**, a Heckman selection model (Heckman, 1979) is used to identify the determinants of household participation in non-farm employment and non-farm income, a simultaneous equations model is used to address the simultaneity of non-farm employment and natural extraction, and an IV-Probit model is applied to investigate the impacts of non-farm income on reducing poverty. We follow Foster et al. (1984) to generate three indicators of poverty, namely the poverty headcount ratio, poverty gap, and poverty severity. The diagnostic tests and corresponding econometric models are conducted using STATA 15.

For the **third paper**, to determine the factors affecting household's participation in land consolidation, a translog stochastic frontier production estimation from true random-effects with Mundlak's adjustments (Greene, 2005; Nguyen et al., 2021; Mundlak, 1978) is first employed to estimate the farming efficiency of rice farmers in Vietnam, and a simultaneous equations model is then applied to address the simultaneity of farming efficiency and land consolidation. To examine the association between land consolidation and rural transformation, a seemingly (un)related regression is employed to control for the interdependence between rural transformation variables, namely income from farm and non-farm sources and their shares in total household income. Lastly, a Propensity Score Matching (PSM) integrated with Difference-in-Differences (DD)

method is utilized to evaluate the impacts of land consolidation on crop production and poverty. We use the Simpson's diversification index to capture the situation of land fragmentation and we follow the FGT method (Foster et al., 1984) to calculate three poverty indicators, including the poverty headcount ratio, poverty gap, and poverty severity. The diagnostic tests and corresponding econometric models are conducted using STATA 15.

In the **fourth paper**, a simultaneous equations model is applied to examine the relationship between crop commercialization and mechanization, a household fixed-effects model with control function (CF) approach (Wooldridge, 2015) is used to estimate the effects of crop commercialization on non-farm employment, and an unconditional quantile regression (UQR) model proposed by Firpo et al. (2009) is employed to assess the distributional effects of crop commercialization on smallholders' farm income and total income. We use the precipitation data from the TRMM to generate the IV for the simultaneous equations models and FE with CF models. The diagnostic tests and corresponding econometric models are conducted using STATA 15.

With regard to the **fifth paper**, a household fixed-effects estimation model is used to identify the role of year-round irrigation at village level on irrigation of rural households, a fixed-effects model with IV is employed to examine the effects of irrigation on crop production and to evaluate the impacts of irrigation on household income and poverty. We use two poverty indicators, namely absolute income poverty at PPP\$ 3.20 per capita a day and multidimensional poverty suggested by the World Bank (World Bank, 2020). The diagnostic tests and corresponding econometric models are conducted using STATA 15.

For the **sixth paper**, a Probit model is used to identify the factors affecting the participation of female rural-urban migrants in online marketplaces, an IV approach based on the heteroscedasticity-based identification strategy suggested by Lewbel (2012) is utilized to examine the effects of online market participation on female migrants' welfare, and an unconditional quantile regression (UQR) model proposed by Firpo et al. (2009) is applied to evaluate the distributional effects of online market participation on female migrants' welfare. The diagnostic tests and corresponding econometric models are conducted using STATA 15.

1.4 Data

The data used in all papers of this dissertation come from the “*Thailand - Vietnam Socio-Economic Panel (TVSEP) - Poverty dynamics and sustainable development: A long-term panel project in Thailand and Vietnam*” funded by the German Research Foundation (DFG FOR 756/2). There are three datasets under the TVSEP project. The first and most important dataset includes the information of rural households collected from a structured questionnaire. The rural household questionnaire consists of several components covering household demographic characteristics, livelihood strategies, household income and expenditure, shocks and risks, assets, financial capital, housing conditions. The detailed description of sampling and survey implementation can be found from Do et al. (2022). The second dataset of the TVSEP project is the village data collected from a structure questionnaire. The village heads were often the ones who answered the questionnaire. The village data include a wide range of information such as infrastructure, local institutions, economic and environmental conditions. The description of the village dataset is explained by Do (2023). The household and village data are used for empirical analyses in the **first, second, third, fourth, and fifth paper**. The last dataset of the TVSEP project is the tracking migrant data. In 2018, a migrant tracking survey was conducted to collect information from rural-urban migrants. The TVSEP project relied on the migration information from the rural household survey in 2017 to identify these rural-urban migrants in cities. The detailed description and survey instrument of the data are explained by Nguyen and Do (2022). The **sixth paper** uses the data from the migrant tracking survey.

The implementation of the new rural development (NRD) program is important in this dissertation. The data from the 2016 wave of the rural, agricultural, and fishery census conducted by the General Statistics Office of Vietnam (GSO) are used to capture the implementation results of the rural development program. This dataset was used in the **first paper**.

The precipitation data from the Tropical Rainfall Measuring Mission (TRMM) which is a joint mission of the National Aeronautics and Space Administration (NASA) and the Japan Aerospace Exploration Agency (JAXA) are used to generate instrumental variables in our empirical analyses. The precipitation data from TRMM are highly spatial with $0.25^\circ \times 0.25^\circ$ resolution and are temporal with daily and 3-hourly records. The rainfall data from the TRMM project are only available for the period from 1998 to 2014.

The precipitation data are used in the **first, second, fourth, and fifth paper**.

1.5 Results

In the **first paper**, it appears that more risk-averse households tend to accumulate more savings as an absorptive capacity and diversify their income sources as an adaptive capacity. Under uncertainty, there is a positive demand for savings and income diversification as a precautionary strategy to cope with unexpected events or to adapt to shocks (Arslan et al., 2018; Yang et al., 2021). In addition, achieving a new rural standard in terms of planning and irrigation positively and significantly increases the absorptive capacity of rural households. We also find that savings and income diversification have a negative and significant influence on reducing consumption due to shocks, which is consistent with the findings of Smith and Frankenberger (2018). Our findings indicate that accumulating savings as an absorptive capacity and diversifying income sources as an adaptive capacity of household resilience significantly help rural households to be better prepared *ex ante* to mitigate the negative effects of shocks in the *ex post* period.

With regard to the **second paper**, besides household characteristics, characteristics of village's infrastructure such as the share of households with cable internet at home and a higher quality of roads have significant and positive effects on household participation in non-farm employment and their income from non-farm activities. These findings indicate that investment in infrastructure and internet development play a significant role in providing rural households with opportunities to participate in non-farm employment and, at the same time, improve their income by encouraging rural self-employment and attracting enterprises to rural regions (Haggblade et al., 2010). Further, income from non-farm activities can significantly reduce poverty in relative terms, absolute terms, as well as in terms of various aspects of poverty such as headcount ratio, poverty gap, and poverty severity.

The results from the **third paper** show that farming efficiency positively and significantly affects the participation of rice farmers in land consolidation. Two variables representing village's infrastructure conditions, namely better road quality and the number of enterprises with at least nine employers, have a negative and significant influence on farmers' participation in land consolidation. These village variables reflect the availability of opportunities for non-farm employment and these opportunities can discourage rice farmers to conduct land consolidation and encourage them to participate

in non-farm activities (Do et al., 2022). Besides, land consolidation positively affects the share of farm income, but it negatively influences the share of non-farm income which imply an important role of land consolidation in stimulating rural transformation. Finally, participation in land consolidation can prevent rice farmers from falling into poverty which is consistent with previous studies (e.g., Wang et al. (2021)).

In the **fourth paper**, we find that crop commercialization is positively and significantly related to mechanization costs in crop production. This finding is consistent with the findings of Liu et al. (2020) and Pingali and Rosegrant (1995) which indicate that crop commercialization positively influences mechanization and, at the same time, crop mechanization positively affects crop commercialization (Owusu and İşcan, 2021). Furthermore, our results also point to the fact that improving crop commercialization positively and significantly affects non-farm income per capita. Crop commercialization, in relative terms, brings higher benefits to smallholder farmers in the poorest groups (i.e., in the 10th and 25th income quantile groups). Therefore, crop commercialization can play a significant role in improving smallholders' income and reducing income inequality.

From the **fifth paper**, we can see that the exposure to weather shocks negatively and significantly influences the share of irrigated land area. This result is in line with the findings of Kaini et al. (2021) and Malek et al. (2018). On the other hand, having year-round irrigation in the village is positively and significantly correlated with the share of household irrigated land area. While weather shocks negatively affect farming efficiency, a higher share of irrigated land increases farming efficiency. Our results are consistent with the findings of Nguyen et al. (2022a) and Kang et al. (2021). In addition, we also find that the lagged variable of the share of irrigated land area negatively and significantly impact both absolute poverty at PPP\$ 3.20 and multidimensional poverty.

In the **sixth paper**, it appears that participation in online business significantly increases the consumption of female migrants and reduces relative consumption poverty. The results from the estimations on the number of activities show that only participating in all three online market activities can have positive and significant impacts on female migrants' consumption and negative and significant effects on the poverty status. When we consider each activity separately, the participation in online market places has a positive and significant effect on female migrants in the higher consumption quantile groups. In particular, participating in online financial transaction positively and significantly affects the consumption of female migrants from the 50th and 75th

expenditure quantile groups. Participating in online trading activities only has a significant effect on female migrants in the 90th expenditure quantile group and the effect of participating in online business is significant for female migrants in the 50th, 75th and 90th groups. These results imply that online market participation is benefiting the better-off, rather than the poor.

1.6 Conclusion and policy implications

Based on the empirical evidence on how household capital and village conditions, especially infrastructure development, can improve rural household resilience capacities, increase income, mitigate the effects of shocks, reduce poverty, and ultimately promote structural change in Vietnam and Southeast Asia in general, several policy implications are proposed to promote structural transformation.

In the **first paper**, the results show that the number of enterprises in the village, strategic planning of socio-economic development, and an adequate irrigation system in the village positively affect the absorptive and adaptive capacity of rural households. We recommend that rural development policies in developing countries should put more emphasis on creating more income-generating activities and employment opportunities, for example by attracting more enterprises to rural villages, developing strategic planning for socio-economic development, or developing an irrigation system at village level. In addition, the local governments should also focus on supporting households in disadvantaged groups, such as those with female heads, having high dependency ratios, relying more on only farming activities, and locating in villages far from provincial centers, to deal with unexpected events. Last, rural development programs could invest more in developing transportation infrastructure (e.g., paved roads) and information and communication technology (ICT) infrastructure (e.g., the internet connectivity).

The findings from the **second paper** imply that rural development policies on creating more non-farm opportunities play an important role in helping rural households increase their income and welfare, and, at the same time, decrease the exploitation of natural resources in developing countries. In particular, investments in developing transportation infrastructure and ICT facilities can provide rural households with better opportunities to participate in non-farm activities. Last, the promotion of education in rural areas can be also important on reducing natural extraction and, thus, improving the sustainability of local communities.

In the **third paper**, the results point to the fact that land consolidation should be stimulated to address several issues of rice farmers such as low income and poverty. Public policy should focus on facilitating the redistribution of agricultural land from farmers who want to give up farming to farmers who want to continue working in agriculture in order to reduce the negative impacts of land fragmentation and accelerate rural transformation. In particular, at the household level, farmers who are more efficient in farming should be encouraged and facilitated to enlarge their farm size by land consolidation. At the village level, rural development policies aimed at stimulating rural transformation should invest more in rural infrastructure (e.g., better road quality and internet availability) and attract more enterprises to rural areas.

In the **fourth paper**, our results show that crop commercialization should be stimulated to accelerate structural change and, at the same time, it can also be the target of interventions in rural development policies to fight against poverty and income inequality. The focus can be placed on supporting the use of mechanization, the improvement of smallholder collective organizations, and the enlargement of farmland. Promoting investments in rural infrastructure such as improving road quality (e.g., paved roads) can provide smallholders with opportunities to engage more in commercializing their products and also to participate in non-farm activities with a better income.

Based on the findings from the **fifth paper**, we can see that the increase of water scarcity induced by climate-driven changes and the increasing demand for water from other economic sectors are posing significant risks to agricultural production, poverty reduction, and food security at the global level. To ensure more efficient farming and improve farmers' income in rural areas, we recommend that governments in developing countries should pay more attention to irrigation development and water management to increase the sustainability and availability of water for irrigation. Given the fact that farmers are increasingly confronted with weather calamities, access to year-round irrigation can solve these problems better than access to irrigation alone. Further, the development of irrigation should go hand in hand with the development of transportation and ICT infrastructure. Together with better irrigation, better rural roads and ICT facilities can significantly improve incomes and reduce poverty among rural households.

Our findings from the **sixth paper** suggest that the governments in developing countries should support rural-urban migration, especially female migrants since they are more vulnerable in urban cities. The support could be placed on facilitating rural-urban

migrants to be able to fully participate in online market activities. The facilitation can be done through improving education and increasing access to the internet in rural areas. Even if access to the internet and online marketplaces is improved, interventions may still be needed to support female migrants in the poorest segment to improve their welfare, as they are the least likely to benefit from ICT development. Therefore, they will not be left behind.

1.7 Outline

Six papers are organized in the following six chapters. Chapter 2 contains the **first paper** “*The role of savings and income diversification in households’ resilience strategies: Evidence from rural Vietnam*” that was published in *Social Indicators Research* in 2023. Manh Hung Do is the single author of the first paper.

The **second paper** titled “*Non-farm employment, natural resource extraction, and poverty: Evidence from household data for rural Vietnam*” is included as Chapter 3 that was published in *Environment, Development and Sustainability* in 2022. In this paper, Manh Hung Do was responsible for formulating ideas, preparing data, creating models, performing formal analysis, writing and editing original draft and revision. Trung Thanh Nguyen and Ulrike Grote were responsible for formulating ideas, proposing methods, writing and editing original draft and revision, and supervision. George E. Halkos was responsible for formulating ideas and editing original draft and revision.

Chapter 4 contains the **third paper** “*Land consolidation, rice production, and agricultural transformation: Evidence from household panel data for Vietnam*” that was published in *Economic Analysis and Policy* in 2023. In this paper, Manh Hung Do was responsible for formulating ideas, preparing data, creating models, performing formal analysis, writing and editing original draft and revision. Trung Thanh Nguyen and Ulrike Grote were responsible for formulating ideas, proposing methods, writing and editing original draft and revision, and supervision.

The **fourth paper**, structured as Chapter 5, is the paper “*Crop commercialization, structural change and income inequality: Insights from smallholder farmers in rural Southeast Asia*” that is currently under review in *World Development*. In this paper, Manh Hung Do was responsible for formulating ideas, preparing data, creating models, performing formal analysis, writing and editing original draft and revision. Trung Thanh Nguyen and Ulrike Grote were responsible for formulating ideas, writing and editing

original draft and revision, and supervision.

The **fifth paper** titled “*Weather shocks, irrigation development and poverty: evidence from rural Northeast Thailand and Central Vietnam*” was published in the *International Journal of Water Resources Development* in 2024. In this paper, Manh Hung Do was responsible for formulating ideas, preparing data, creating models, performing formal analysis, writing and editing original draft and revision. Trung Thanh Nguyen was responsible for formulating ideas, writing and editing original draft and revision, and supervision.

Chapter 6 contains the **sixth paper** “*Female rural-urban migrants and online marketplaces in emerging economies: Evidence from Thailand and Vietnam*” that was published in the *Asia & the Pacific Policy Studies* in 2022. In this paper, Manh Hung Do was responsible for formulating ideas, preparing data, creating models, performing formal analysis, writing and editing original draft and revision. Trung Thanh Nguyen was responsible for formulating ideas, writing and editing original draft and revision, and supervision.

Table 1.1. Overview of papers

	Title	Author(s)	Papers' history
Chapter 2	The role of savings and income diversification in households' resilience strategies: Evidence from rural Vietnam	Do, M. H.	Published in <i>Social Indicators Research</i> 2023, 168(1-3), 353-388.
Chapter 3	Non-farm employment, natural resource extraction, and poverty: Evidence from household data for rural Vietnam	Do, M. H. , Nguyen, T. T., Halkos, G., Grote, U.	Published in <i>Environment, Development and Sustainability</i> 2022. Early version presented at: the 6 th Economics of Natural Resources and the Environment Conference (ENVECON), 11-12 June 2021.
Chapter 4	Land consolidation, rice production, and agricultural transformation: Evidence from household panel data for Vietnam.	Do, M. H. , Nguyen, T. T., Grote, U.	Published in <i>Economic Analysis and Policy</i> 2023, 77, 157-173.
Chapter 5	Crop commercialization, structural change and income inequality: Insights from smallholder farmers in rural Southeast Asia	Do, M. H. , Nguyen, T. T., Grote, U.	Under review in <i>World Development</i> Early version presented at: the 9 th Economics of Natural Resources and the Environment Conference (ENVECON), 8-9 December 2023.
Chapter 6	Weather shocks, irrigation development and poverty: evidence from rural Northeast Thailand and Central Vietnam	Do, M. H. , Nguyen, T. T.	Published in <i>International Journal of Water Resources Development</i> 2024, 40(3), 463-486.
Chapter 7	Female rural-urban migrants and online marketplaces in emerging economies: Evidence from Thailand and Vietnam	Nguyen, T. T., Do, M. H.	Published in <i>Asia & the Pacific Policy Studies</i> 2022, 9(3), 317-342.

During the PhD period, the author was also involved as the main author or co-author in six published papers that are not included as part of the dissertation:

1. Nguyen, T. T., **Do, M. H.**, Rahut, D. B., Nguyen, V. H., and Chhay, P. (2023). Female leadership, internet use, and performance of agricultural cooperatives in Vietnam. *Annals of Public and Cooperative Economics*, 94(3), 877–903. <https://doi.org/10.1111/apce.12434>
2. Wang, M., and **Do, M. H.** (2023). Reported shocks, households' resilience and local food commercialization in Thailand. *Journal of Economics and Development*, 25(2), 153-170. <https://doi.org/10.1108/JED-10-2022-0204>
3. Nguyen, T. T., Nguyen, T. T., **Do, M. H.**, Nguyen, D. L., and Grote, U. (2022). Shocks, agricultural productivity, and natural resource extraction in Rural Southeast Asia. *World Development*, 159, 106043. <https://doi.org/10.1016/j.worlddev.2022.106043>
4. Nguyen, T. T., **Do, M. H.**, and Rahut, D. (2022). Shock, risk attitude and rice farming: evidence from panel data for Thailand. *Environmental Challenges*, 6, 100430. <https://doi.org/10.1016/j.envc.2021.100430>
5. **Do, M. H.**, and Park, S. C. (2022). Factors affecting foreign direct investment and economic performance in developing economies: Evidence from Vietnam. *International Journal of Public Administration*, 45(1), 64-80. <https://doi.org/10.1080/01900692.2021.1995745>
6. Nguyen, T. T., and **Do, M. H.** (2021). Impact of economic sanctions and counter-sanctions on the Russian Federation's trade. *Economic Analysis and Policy*, 71, 267-278. <https://doi.org/10.1016/j.eap.2021.05.004>

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CHAPTER 2: THE ROLE OF SAVINGS AND INCOME DIVERSIFICATION IN HOUSEHOLDS' RESILIENCE STRATEGIES: EVIDENCE FROM RURAL VIETNAM

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CHAPTER 3: NON-FARM EMPLOYMENT, NATURAL RESOURCE EXTRACTION, AND POVERTY: EVIDENCE FROM HOUSEHOLD DATA FOR RURAL VIETNAM

This chapter is a modified version of the article:

“Non-farm employment, natural resource extraction, and rural household’s welfare: Evidence from panel data for Vietnam”. This paper was presented at the 6th Economics of Natural Resources and the Environment Conference (ENVECON), 11-12 June 2021, held online and organized by the University of Thessaly, Volos, Greek.

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CHAPTER 4: LAND CONSOLIDATION, RICE PRODUCTION, AND AGRICULTURAL TRANSFORMATION: EVIDENCE FROM HOUSEHOLD PANEL DATA FOR VIETNAM

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<https://doi.org/10.1016/j.eap.2022.11.010>

CHAPTER 5: CROP COMMERCIALIZATION, STRUCTURAL CHANGE AND INCOME INEQUALITY: INSIGHTS FROM SMALLHOLDER FARMERS IN RURAL SOUTHEAST ASIA

This chapter is a modified version of the paper:

“Smallholders’ crop commercialization and agriculture’s structural change in Southeast Asia”. This paper was presented at the 9th Economics of Natural Resources and the Environment Conference (ENVECON), 8-9 December 2023, held online and organized by the University of Thessaly, Volos, Greek.

Under review in

World Development

Abstract

Given the importance of understanding the impact of crop commercialization on structural change in agriculture and equality at the micro level, we examine the effects of smallholder crop commercialization on structural change and assess how the impacts of crop commercialization on smallholder income are distributed across income quantiles. To address these research issues, we use a balanced panel of 2,867 rural smallholder farmers collected in three survey waves from two middle-income countries in Southeast Asia. To examine the impact of crop commercialization on structural change, our simultaneous equation model results show that crop commercialization is positively related to crop mechanization. In addition, the results of fixed-effects estimation with a control function approach indicate that crop commercialization has a positive influence on non-farm employment. To assess the distributional impact of crop commercialization on income, the results of unconditional quantile regression models suggest that smallholders in the 10th and 25th quantiles benefit most, relatively speaking, from crop commercialization. Hence, we recommend facilitating the commercialization of crops by smallholder farmers to stimulate the structural transformation of agriculture at the micro level and to reduce income inequality of smallholder farmers.

Keywords: Panel data; simultaneous equation; fixed-effects with control function; income inequality.

JEL code: C33; Q00; Q12.

5.1 Introduction

Structural change in agriculture is an integral part of economic development in developing economies (Amare et al., 2023; Barrett et al., 2017; Pingali and Abraham, 2022). The process of structural change is accompanied by adjustments within agriculture and outside the sector. Within the sector, the process starts with the shift from subsistence to commercial production systems (Barrett, 2008; Von Braun, 1995), increased productivity and mechanization (Gollin et al., 2002; Pingali, 1997; Timmer, 1988; Üngör, 2013), and increasing land consolidation (Do et al., 2023; Nguyen et al., 2021). Outside the sector, the development of non-farm sectors induced by economic growth provides employment opportunities which enable the shift of labor from agriculture (low productivity) to non-farm sectors (more productive) (Hansen and Prescott, 2002; Pingali and Abraham, 2022). Structural change leads to income growth and economic development (McMillan et al., 2014; Pingali et al., 2019; Timmer, 2017).

The commercialization of agriculture underpins economic structural change in many developing countries (Johnston and Kilby, 1975; Lipton, 2006; Sitko and Jayne, 2014). Nevertheless, there are unbalanced studies between macro and micro levels (Amare et al., 2023). On the one hand, the macro-level literature has shown ample evidence of structural changes at national, regional, and sectoral levels (e.g., Adamopoulos and Restuccia, 2014; Barrett et al., 2017; Bustos et al., 2016; Bustos et al., 2020; Diao et al., 2017; McArthur and McCord, 2017; Teignier, 2018). These macroeconomic studies have covered multidimensional aspects of agriculture and structural change in developing countries. On the other hand, there are fewer microeconomic studies at farm or smallholder levels which focus separately on either commercialization (e.g., Carletto et al., 2017; Ochieng et al., 2020; Ogutu et al., 2020; Schulte et al., 2023) or structural change (e.g., Amare et al., 2023; Bachewe et al., 2018; Diao et al., 2019; Liu et al., 2020). Hence, the question “How can smallholder commercialization affect structural change?” remains unanswered, even though smallholder farmers, who decide what to grow and how much to sell, make a significant contribution to global food systems, particularly in the Global South. They produce 80% of the food in Asia and Africa (Abraham et al., 2022; Lowder et al., 2016). Furthermore, understanding structural change at the micro level could be crucial for stimulating structural change at the macro level (Amare et al., 2023).

Previous studies have shown a heterogeneous effect of commercialization on the well-being of different groups of participants (Ochieng et al., 2020; Ogutu et al., 2020; Schulte et al., 2022). The unbalanced effects of commercialization have been noted in some studies

(Abraham et al., 2022; Tabe Ojong et al., 2022), for example, significant effects on dietary diversity for lower income groups (Chegere and Kauky, 2022), larger effects on income for better-off households (Ogutu and Qaim, 2019), and higher gains for asset-rich households (Tabé Ojong et al., 2022). Interestingly, although commercialization is directly related to small-scale farming, there is little evidence on how the commercialization of agricultural products (e.g. crops), heterogeneously affects farm income and household income. Given this lack of empirical evidence on the unequal effects of commercialization, it is worthwhile to examine the distributional effects of crop commercialization on farm income and the total income of smallholder farmers.

Against this background, we aim to answer two key questions, including (i) Can smallholder-led crop commercialization affect structural change in agriculture at micro level? And (ii) How are the benefits of crop commercialization distributed across different quantile groups of farm income and total income of smallholder farmers? We address the first question by examining the impact of micro-scale commercialization of crops on the adoption of mechanization and the shift to non-farm employment. These indicators are often used as indicators of structural change (Amare et al., 2023; Barrett et al., 2017; Diao et al., 2017; Do et al., 2023; Wang et al., 2016). In addition, we consider the simultaneity of commercialization and mechanization in crop production by using a simultaneous equations model. With regard to the second question, we employ an unconditional quantile regression (UQR) model to investigate who benefits more from crop commercialization in terms of farm income and total income. We use the panel data for Thailand and Vietnam, two emerging economies in Southeast Asia, to examine these issues for several reasons. First, Thailand and Vietnam are among the fast-growing economies and are still on their way to higher positions on the income ladder of middle-income economies (Pingali and Abraham, 2022; Klasen and Waibel, 2015). The process of structural change in agriculture and non-agriculture is not yet complete. Second, although these countries are experiencing rapid economic growth, the vast majority of the rural population are smallholders who own less than 2 hectares of land and rely on farm income (Do and Nguyen, 2023; Lowder et al., 2016; Nguyen et al., 2022). Finally, Thailand and Vietnam are among the major food exporters in Asia, and studying the food systems in these countries can have important implications for food security on a global scale.

The remaining parts of this article are divided into five sections. Section 2 presents the conceptual framework and literature review. Section 3 describes the study sites and data. Section 4 explains the research method. Section 5 analyses and discusses the results. Section 6 summarizes the findings and suggests policy implications.

5.2 Conceptual framework and literature review

A smallholder farmer in developing countries may have different livelihood strategies, which can be explained by the sustainable livelihood framework proposed by Ashley and Carney (1999) and also applied by Nguyen et al. (2015) and Do et al. (2022) (see Figure 5.1). This framework starts with the smallholder livelihood platforms, which include their capital such as human capital (e.g., education, household size, and household labor), social capital (e.g., membership in social organizations and social networks), natural capital (e.g., farm land), physical capital (e.g., equipment and machinery that the smallholders can use for their livelihoods such as phones and vehicles), and financial capital (e.g., smallholders' wealth such as accumulated asset values and their surrounding environment (e.g., village) which is characterized, for example, by the availability of markets, infrastructure for transport and communication (e.g., roads, phone line, and the internet), and non-farm employment opportunities (e.g., non-farm enterprises or firms in the village).

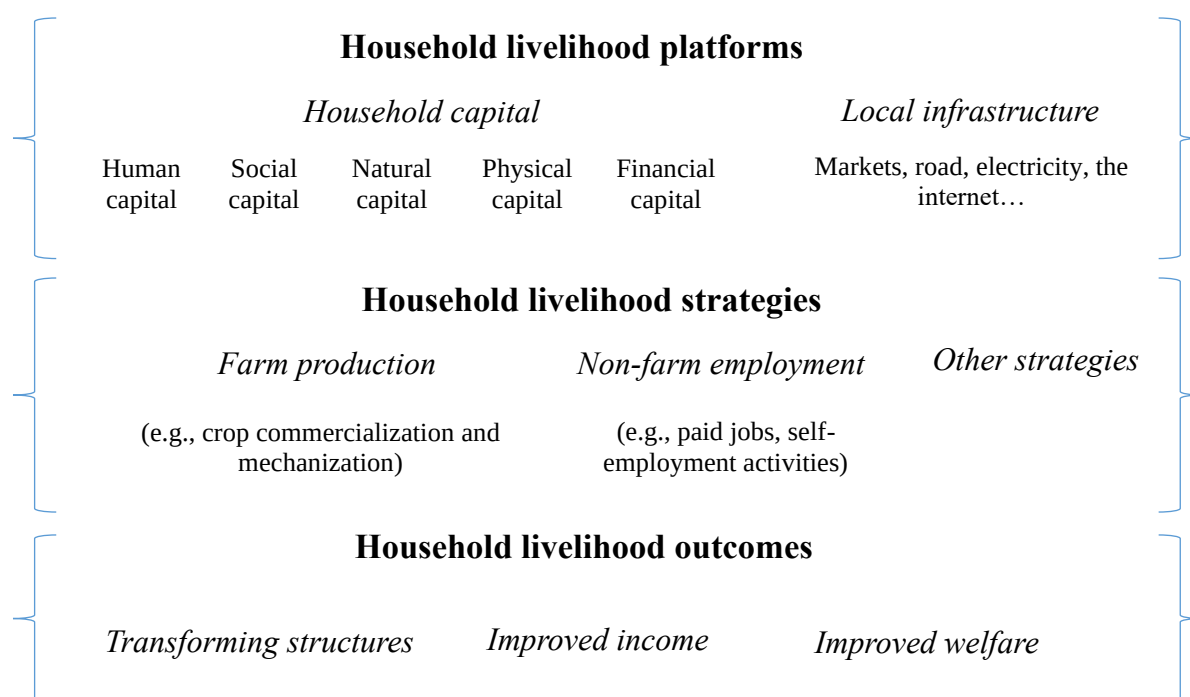


Figure 5.1. Conceptual framework of household livelihood analysis (modified from Ashley and Carney, 1999; Nguyen et al., 2015; Do et al., 2022).

With the help of these livelihood platforms, the smallholder can pursue various livelihood strategies to generate earnings. For instance, this smallholder may stick to farm production such as crop, livestock, or aquaculture production. Alternatively, the smallholder can also diversify income into non-farm sources such as wage employment or self-employment. While agricultural activities are usually selected by default, participation in non-farm activities

requires certain conditions at both the household level (e.g., household education and size) and the village level (e.g., distance to centers, road quality, and infrastructure for information and communication technology (ICT)) (Do et al., 2022; Rahut and Micevska Scharf, 2012; Rao and Qaim, 2011). The shift from farm to non-farm sectors implies the change of household's income or production structure (i.e., structural change) (Gollin et al., 2002; Timmer, 1988). Other strategies such as the extraction of natural resources can also be pursued by the smallholder.

Regarding farm production, the decision of the smallholder on how much to sell or keep for own consumption/use (i.e., the rate of commercialization) is part of households' livelihood strategies to improve income (Ochieng et al., 2020). The rate of commercialization can be measured by the share of sold products in total output (Owusu and İşcan, 2021; Ogutu et al., 2020; Von Braun, 1995). The extent of commercialization is not only about the production and specialization of cash crops, but can also relate to staple crops, of which the smallholder can sell a part and keep the rest for own consumption/use (Ochieng et al., 2020; Schulte et al., 2023).

The commercialization of crop output (as the main product of agriculture) triggers structural change (Barrett, 2008; Pingali, 1997). Changing structures also include the development of farm mechanization (within the sector) and the shift to non-farm activities (outside the sector) (Amare et al., 2023). This structural change leads to income growth and poverty reduction (McMillan et al., 2014; Pingali et al., 2019; Timmer, 2017). At this point the question arises as to whether promoting smallholder's commercialization can contribute to stimulating structural change and have an impact on their welfare.

There are two different strands of literature on commercialization and structural change at the smallholder level. The first strand focused on the determinants and impacts of commercialization. Empirical evidence on the determinants of smallholder commercialization has shown that demographic characteristics (i.e., human capital), farm/land characteristics, and non-farm employment are important driving factors (Owusu and İşcan, 2021; Woldeyohanes et al., 2017). The impacts of commercialization have been intensively studied in various aspects such as productivity, income, consumption, asset growth, poverty, and especially nutrition in smallholder households (Abraham et al., 2022; Carletto et al., 2017; Cazzuffi et al., 2020; Ochieng et al., 2020; Ogutu et al., 2020; Schulte et al., 2023). The other strand of literature dealing with structural change has shown that an increase in smallholder income, better education of smallholders, greater proximity to cities and better local infrastructure have a significant influence on structural change (Amare et al., 2023; Bachewe et al., 2018; Diao et al., 2019; Liu et al., 2020; Pinstруп-Andersen and Shimokawa, 2006).

However, there are several research gaps in study design, methodology and empirical evidence that require further attention. First, regarding study design, there are very few studies that use smallholder data, although smallholders play a vital role in global food systems and are vulnerable to poverty (Amare et al., 2023; Collier and Dercon, 2014; Liu et al., 2020; Lowder et al., 2016). More importantly, of the previous studies on commercialization, only 42% used quantitative methods and 87% of these quantitative studies were based on cross-sectional data (Abraham et al., 2022). However, using cross-sectional data may not be sufficient to examine smallholder farmers' livelihoods since these data only capture the information at a single point in time. In this study, we contribute to the current literature by providing empirical evidence from panel data from a long-term project that collected information from smallholders in Thailand and Vietnam, two emerging economies in Southeast Asia.

Second, in terms of methodology, previous studies have not addressed the issues of selection bias, unobservable factors, and endogeneity of smallholder commercialization (Abraham et al., 2022). These issues should be taken into account in impact assessment studies (Barrett and Carter, 2020). Fourth, the relationship between commercialization and mechanization has not been examined in previous studies. Mechanization is an indicator of structural change (Amare et al., 2023; Liu et al., 2020). The shift from subsistence to commercial production, triggering structural change, promotes the mechanization of agricultural production as labor is shifted to the non-farm sector (Pingali and Rosegrant, 1995; Pingali, 1997; Timmer, 1988). At the same time, mechanization is an important driving factor for crop commercialization and affects the market supply decision of smallholders (Nguyen and Warr, 2020; Owusu and İşcan, 2021; Woldeyohanes et al., 2017). These details suggest the potential simultaneity of crop commercialization and mechanization (Carletto et al., 2017). We offer an attempt to account for this problem by using a simultaneous equations model.

Finally, in terms of empirical evidence, there is no evidence of the distributional impacts of commercialization on smallholders' farm income, although commercialization is directly linked to income from farming. Furthermore, there are concerns about the heterogeneous effects of commercialization on the welfare of different participant groups. The available evidence shows that commercialization appears to benefit wealthier participants more than those who are worse-off (Ogutu and Qaim, 2019; Tabe Ojong et al., 2022). In this regard, commercialization can widen the gap between the rich and poor, ultimately exacerbating the problem of income inequality. However, the findings to date are mainly based on the absolutely heterogeneous effects of commercialization. We contribute to the literature by using a more appropriate approach to estimate the relative distributional effects of commercialization on smallholders'

farm income and total income. Overall, our results are expected to significantly contribute to the literature on smallholder commercialization and structural change in developing countries.

5.3 Data and descriptive summary

5.3.1 Study sites and data

We use data from the “Thailand - Vietnam Socio-Economic Panel (TVSEP): Poverty dynamics and sustainable development: A long-term panel project in Thailand and Vietnam” funded by the German Research Foundation (DFG FOR 756/2) (see www.tvsep.de for more information about the project). The TVSEP project aims to provide a better understanding of income and poverty-vulnerability dynamics in rural areas of Thailand and Vietnam, two emerging economies in Southeast Asia (Hardeweg et al., 2013). The sampling process under TVSEP followed the guidelines of the United Nations Department of Economic and Social Affairs (UN, 2005). The sampling process began with the selection of three provinces in Thailand (Buriram, Nakhon Phanom, and Ubon Ratchathani) and three provinces in Vietnam (Ha Tinh, Thua Thien Hue, and Dak Lak) (see Figure 5.2). The main reasons for selecting these provinces were that they had suitable conditions for assessing vulnerability to poverty, namely low average per capita income, high dependence on agriculture, the existence of particular risk factors, and poor infrastructure.

A three-stage stratified random sampling process was then used to draw samples. In the first stage, sub-districts (in Thailand) or communes (in Vietnam) were selected. At this stage, the proportion of sub-district’s/commune’s population in the sample districts was used to weight the selection. Two villages were then selected with a probability proportional to their size in the sub-district/commune population. In the final stage, ten households in each village were randomly selected from a list of all households in the sample villages with equal selection probability. This resulted in a total sample of 440 villages and 4,400 households in two countries (one half for each country).

Two structured questionnaires for rural households and villages have been used as survey instruments to collect data as part of the TVSEP. Since 2016, the Computer Assisted Personal Interviewing (CAPI) has been employed on tablets to collect data. The structured rural household questionnaire consists of multiple components covering human characteristics of the households (e.g. age, education, employment and health status), income and livelihood strategies (e.g. crop production, livestock and aquaculture production, wage employment, and non-farm self-employment), household expenditure, shock experience and risks, assets, and

housing conditions. In the household surveys, the reference period is always 12 months (from May of the last year to April of the survey year). This reference period is used to record the data of household income and consumption. Heads of households are often the respondents in rural household interviews. In the structured questionnaire for rural villages, it is always the village heads who provide the information about village characteristics. The village questionnaire captures the information at the village level such as infrastructure development (e.g. road quality, irrigation, access to information and communication technology (ICT), and access to electricity), the availability of enterprises in the villages, and the distance from the villages to important facilities such as the closest markets and to provincial centers (the detailed definition and measurement of variables is presented in Table A1 in the Appendices). To date, the TVSEP surveys, which have been conducted entirely in all six provinces in Thailand and Vietnam, include the waves of 2007, 2008, 2010, 2013, 2016, and 2017. The “random attrition” of the TVSEP data (i.e., the number of households could not be re-interviewed over time) is about 3% per wave (further information on the survey waves can be found at <https://www.tvsep.de/en/survey-documents>).



Figure 5.2. The study sites in Thailand and Vietnam under the Thailand–Vietnam Socio-Economic Panel (Source: Nguyen et al., 2020b).

To construct the sample and extract the data for this study, we focus on the waves three waves including 2010, 2013, and 2016 (the village data was not available for the 2008 and 2017 waves and the household’s asset value was not available for the 2007 wave). The combination

of these waves further yields equal time gaps between waves (every 3 years). Then, we use only identical households (those households were always interviewed in these selected waves) to better examine the process of structural change. As a result, we have balanced panel data of 2,867 households (1,370 households from Thailand and 1,497 households from Vietnam). These extracting processes increase the attrition rate of the sample by approximately 3.9% compared to the number of households in the first wave. Hence, the final sample consists of 8,601 observations from Thailand and Vietnam in three years.

In addition, we use the precipitation data from the Tropical Rainfall Measuring Mission (TRMM), a joint mission of the National Aeronautics and Space Administration (NASA) and the Japan Aerospace Exploration Agency (JAXA). The rainfall data from the TRMM project are spatial with a resolution of $0.25^\circ \times 0.25^\circ$ and temporal with daily and 3-hourly records (see Kummerow et al. (1998) for the sensors and data algorithms of TRMM precipitation data). The data is only available for the period 1998 to 2014.

5.3.2 Description of data

Table 5.1 presents the descriptive summary of household's commercialization, structural change, and income. Panel A in this Table reports the changes in crop commercialization, structural change indicators, and income between 2010 and 2016. There were significant fluctuations of smallholders' crop commercialization in the period 2010 and 2016. The commercialization rate decreased from about 52% in 2010 to 49% in 2013 and increased to over 56% in 2016. The differences in commercialization rates were statistically significant.

When it comes to structural change indicators, all of these indicators show an increasing trend between 2010 and 2016, both in absolute and relative terms. In absolute monetary terms, mechanization costs per hectare (ha) increased from PPP\$ 142 per ha in 2010 to about PPP\$ 224 per ha in 2016. This change was statistically significant. As far as the non-farm income per capita is concerned, there was a significant increase between 2010 and 2016 when it rose from PPP\$ 737 per capita in 2010 to PPP\$ 1011 per capita in 2013, and reached PPP\$ 1203 per capita in 2016. In relative terms, the share of mechanization costs was roughly 18% in 2010, then increased to about 20% and 28% in 2013 and 2016, respectively. These changes were statistically significant over time. On the other hand, the proportion of members working in non-farm employment remained almost unchanged at 21% between 2010 and 2013. However, this share jumped to 31% in 2016, making the difference between 2013 and 2016 highly significant.

Table 5.1. Descriptive summary of households' commercialization, structural change, and income.

	2010 (n =2867)	2013 (n =2867)	2016 (n =2867)	Mean difference		
				2010 vs. 2013	2013 vs. 2016	2010 vs. 2016
Commercialization of crop products (%)	52.21 (29.84)	48.67 (32.21)	56.09 (30.55)	-3.53***, a	7.41***, a	3.88***, a
Share of mechanization costs in total production costs (%)	17.90 (17.37)	19.52 (19.10)	28.44 (22.63)	1.62***, a	8.92***, a	10.55***, a
Mechanization costs per hectare (PPP\$)	142.40 (313.53)	154.44 (583.53)	224.65 (655.84)	12.05 ^a	70.21***, a	82.26***, a
Share of non-farm laborers (%)	21.41 (26.81)	21.67 (26.64)	31.20 (31.12)	0.26 ^a	9.53***, a	9.79***, a
Non-farm income per capita (PPP\$)	736.84 (1594.49)	1011.24 (4482.52)	1202.88 (4128.27)	274.40***, a	191.63*, a	466.03***, a
Daily farm income per capita (PPP\$)	1.83 (2.90)	2.42 (4.87)	3.29 (10.58)	0.60***, a	0.87***, a	1.47***, a
Daily household income per capita (PPP\$)	4.87 (6.22)	6.00 (15.89)	8.49 (15.97)	1.14***, a	2.49***, a	3.63***, a

Note: Standard deviations in parentheses; ^a: Two-sample t-test; *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Important details emerge from the descriptive statistics of smallholders' farm income and total income. First, both the daily farm income per capita and daily household income per capita variables showed an upward trend in the period of 2010 and 2016, and the changes over time were all significant. In particular, the values of daily farm income per capita in 2010, 2013, and 2016 were about PPP\$ 1.83, PPP\$ 2.42, and PPP\$ 3.29. In a similar trend, daily household income per capita of smallholder farmers increased from PPP\$ 4.87 in 2010 to PPP\$ 6.00 in 2013, reaching a peak of PPP\$ 8.49 in 2016. Second, farm income still played a significant role in the livelihoods of smallholder farmers, accounting for around 40% of total income over time. This implies that agriculture continued to be an important source of income for smallholders in our study sites. The improvement in farm income came despite a decline in crop commercialization between 2010 and 2013. This implies that crop commercialization could potentially have unequal effects on farm income and total income of smallholders. In fact, Figure 5.3 depicts the distribution of the share of crop commercialization across farm income and total income quartiles. We could see that there was a significant gap in the crop commercialization ratio between the first quartile group and the last quartile group of farm income quartiles (about 40% in the first quartile group versus more than 60% in the last quartile). This gap between the two groups was about 20% over time. Besides, in 2013, there was a slight decline in the share of crop commercialization of the smallholder farmers in the first 25%, 50%, and 75% farm income quartiles, while the share of the last 25% remained almost unchanged.

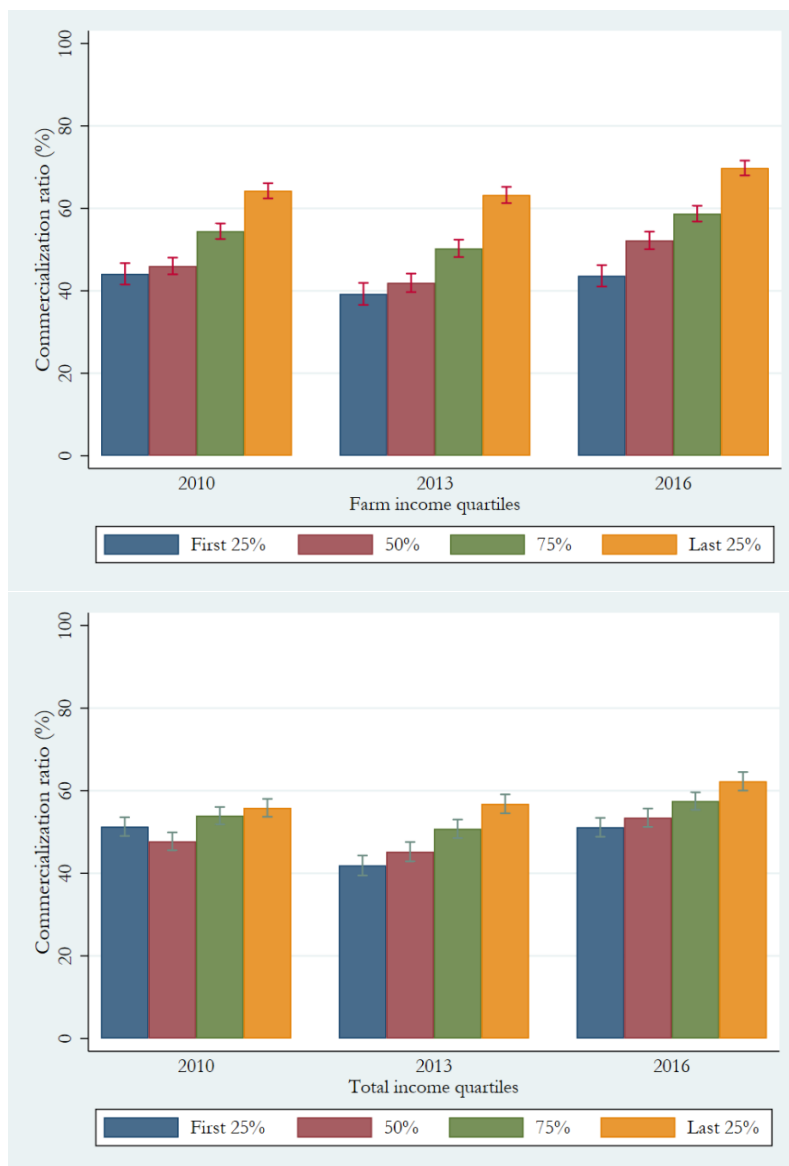


Figure 5.3. Distribution of crop commercialization ratio across farm and total income quartiles.

Panel A1 of Table 5.2 describes the human capital of the smallholders in our sample. It turned out that the average age of household heads was around 55 years. Most household heads were male, but there was a downward trend between 2010 and 2016 (from 82% in 2010 to 80% in 2013 and to 77% in 2016). The difference of this change was significant over time. While the average years of schooling for household heads remained almost unchanged over time, the average years of schooling for adult household members showed a decreasing trend. In particular, adult members' average years of schooling fell from 6.23 in 2010 to 5.64 in 2013, and reached a low at 5.41 in 2016. The average number of nucleus members of households in our sample was four, but the average number of members decreased to 3.89 in 2016. The share of laborers showed an increasing trend from 2010 to 2016. In 2010, the share was about 70%, then rose to 72% in 2013 and peaked at 81% in 2016. The change in the share of laborers was statistically significant over time.

Table 5.2. Descriptive summary of household capital and village characteristics.

	2010 (n =2867)	2013 (n =2867)	2016 (n =2867)	Mean difference		
				2010 vs. 2013	2013 vs. 2016	2010 vs. 2016
<i>A. Household capital</i>						
<i>A1. Human capital</i>						
Age of head (years)	52.64 (12.63)	55.27 (12.44)	57.26 (11.94)	2.63***, a	1.99***, a	4.62***, a
Gender of head (male = 1)	0.82 (0.39)	0.80 (0.40)	0.77 (0.42)	-0.02**, b	-0.03***, b	-0.05***, b
Education of head (years of schooling)	5.92 (3.45)	5.92 (3.48)	5.99 (3.48)	0.01 ^a	0.06 ^a	0.07 ^a
Education of adult members (years of schooling)	6.23 (2.42)	5.64 (2.58)	5.41 (2.76)	-0.59***, a	-0.23***, a	-0.82***, a
Household size (persons)	4.34 (1.69)	4.14 (1.67)	3.89 (1.60)	-0.19***, a	-0.25***, a	-0.44***, a
Share of laborers (%)	70.60 (22.43)	72.84 (22.55)	81.00 (21.92)	2.24***, a	8.16***, a	10.40***, a
<i>A2. Social capital</i>						
Member of PSO (yes = 1)	0.42 (0.49)	0.42 (0.49)	0.40 (0.49)	-0.00 ^b	-0.01 ^b	-0.02 ^b
Ethnicity of head (majority = 1)	0.85 (0.36)	0.84 (0.37)	0.85 (0.36)	-0.01 ^b	0.01 ^b	0.00 ^b
Religion of head (Buddhist = 1)	0.50 (0.50)	0.51 (0.50)	0.51 (0.50)	0.01 ^b	-0.01 ^b	0.01 ^b
<i>A3. Natural capital</i>						
Total land area (hectares)	2.42 (3.64)	2.70 (3.82)	2.05 (2.24)	0.28***, a	-0.65***, a	-0.38***, a
All land plots in the same village (yes = 1)	0.75 (0.43)	0.56 (0.50)	0.63 (0.48)	-0.19***, b	0.07***, b	-0.12***, b
<i>A4. Physical and financial capital</i>						
Number of phones	1.43 (1.25)	2.10 (1.43)	1.58 (1.13)	0.66***, a	-0.52***, a	0.14***, a
Number of large transport vehicles	0.14 (0.44)	0.19 (0.50)	0.23 (0.60)	0.05***, a	0.04***, a	0.09***, a
Asset value per capita (PPPS)	1103.68 (2092.02)	1619.52 (3524.36)	1850.65 (3448.96)	515.84***, a	231.13**, a	746.98***, a
<i>B. Village characteristics</i>						
Number of enterprises	0.10 (0.51)	0.24 (1.01)	0.29 (1.20)	0.14***, a	0.05 ^a	0.19***, a
Having paved roads instead of dirt roads (yes = 1)	0.76 (0.43)	0.79 (0.41)	0.92 (0.27)	0.03**, b	0.13***, b	0.16***, b
Share of households with access to electricity (%)	98.47 (6.29)	97.98 (9.49)	98.86 (6.35)	-0.49**, a	0.88***, a	0.40**, a
Share of households with phone line at home (%)	58.59 (40.92)	93.44 (14.45)	95.84 (10.57)	34.85***, a	2.40***, a	37.25***, a
Share of households with cable internet at home (%)	1.58 (4.67)	3.80 (8.28)	6.92 (11.39)	2.22***, a	3.12***, a	5.34***, a
Distance to provincial capital (km)	50.01 (29.36)	48.24 (29.33)	50.21 (31.75)	-1.77**, a	1.98**, a	0.21 ^a
Distance to the closest market (km)	5.95 (6.96)	5.96 (6.96)	5.98 (6.71)	0.01 ^a	0.02 ^a	0.03 ^a

Note: Standard deviations in parentheses; ^a: Two-sample t-test; ^b: Non-parametric rank-sum test; *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Panels A2 and A3 in Table 5.2 contain information on social and natural capital, respectively. It appeared that about 40% of the households in the sample were members of a political and social organization (PSO), about 85% of the household heads belonged to the ethnic majority group (i.e., the Thai majority in Thailand or the Kinh majority in Vietnam), and

about half of the sample was of Buddhist religious affiliation. The changes in social capital showed no significant difference between 2010 and 2016. Regarding household natural capital, smallholders in our sample had an average land size of about 2 hectares (ha). The average area was about 2.42 ha in 2010, then increased to 2.70 in 2013 and decreased to about 2.05 ha in 2016. The change in land size over time was statistically significant. The share of smallholders whose entire land is in the same village fell from 75% in 2010 to 63% in 2016.

Panel A4 in Table 5.2 shows the descriptive summary of physical and financial capital. It appeared that smallholders in our sample tended to own an increasing number of phones and large transport vehicles between 2010 and 2016. The total asset value per capita increased from about PPP\$ 1,104 in 2010 to \$ 1,620 in 2013 and reached PPP\$ 1,851 in 2016. The change in assets per capita over time was statistically significant.

Panel B in Table 5.2 shows the village characteristics in which the smallholders in our sample live. We could see that most village infrastructure indicators showed significant improvement during the period 2010 to 2016. In particular, the average number of enterprises in the villages rose from 0.1 to 0.29 between 2010 and 2016; the number of villages having paved roads instead of dirt roads also increased from 76% to 92% over the same period. In 2016, more than 95% of villagers in our sample had access to electricity and phone lines at home, while less than 10% of them had cable internet at home. It was also found that the average distance from village to the provincial capital and the distance from village to the closest market were about 50 kilometers and 6 kilometers, respectively.

5.4 Research method

5.4.1 Examining the relationship between crop commercialization and mechanization

We begin the empirical strategy by explaining how we examine the interdependence of commercialization and mechanization in smallholder crop production. It is evident that commercialization triggers structural change and promotes the mechanization of agricultural production as the shift of labor from the farm to the non-farm sectors occurs (Pingali, 1997; Timmer, 1988) and it is also an indicator of structural change (Amare et al., 2023; Liu et al., 2020). At the same time, mechanization is an important driving factor of commercialization (Owusu and İşcan, 2021; Schulte et al., 2022) and influences the market supply decision of smallholders (Woldeyohanes et al., 2017). This implies the interrelationship between crop commercialization and mechanization. We address this problem by employing the simultaneous equations model which can be specified as follows:

$$\mathbf{C}_{iw} = \beta_0 + \beta_1 \mathbf{M}_{iw} + \beta_2 \mathbf{R}_{iw} + \beta_3 \mathbf{X}_{iw} + \varepsilon_{iw} \quad (1)$$

$$\mathbf{M}_{iw} = \gamma_0 + \gamma_1 \mathbf{C}_{iw} + \gamma_2 \mathbf{Z}_{iw} + \gamma_3 \mathbf{X}_{iw} + \epsilon_{iw} \quad (2)$$

In Equations (1) and (2), $\mathbf{C}_{iw}$ and $\mathbf{M}_{iw}$ refer to the ratio of sales values to total crop production values and mechanization costs per ha of smallholder i in wave w , respectively. $\mathbf{X}_{iw}$ is a group of control variables at the household and village levels. At the household level, we control for smallholders' capital, namely human capital (i.e., age of head, gender of head, years of schooling of head, mean years of schooling of adult members, household size, and share of laborers), social capital (i.e., membership in PSO, ethnicity of head, and religion of head), natural, physical and financial capital (i.e., total land area, location of land plots, numbers of phones and large transport vehicles, and asset-poor smallholder). At the village level, we take into account important factors of village infrastructure and employment opportunities, namely number of enterprises, road quality, share of households with access to electricity, share of households with phone line at home, share of households with cable internet at home, and distance to provincial capital. ε_{iw} and ϵ_{iw} are the error terms in Equations 1 and 2, respectively.

To account for the endogeneity of crop commercialization and mechanization in estimating Equations (1) and (2), we employ two sets of instrumental variables (IV) as $\mathbf{R}_{iw}$ in Equation 1 and $\mathbf{Z}_{iw}$ in Equation (2). To satisfy the requirement of IVs for simultaneous equations, $\mathbf{R}_{iw}$ should not be correlated with crop mechanization ($\mathbf{M}_{iw}$) and $\mathbf{Z}_{iw}$ should not have a correlation with smallholders' mechanization ($\mathbf{C}_{iw}$). To construct the IVs for $\mathbf{C}_{iw}$ (as $\mathbf{R}_{iw}$ in Equation (1)), we use the distance from village to the closet market as an additional IV to estimate Equation (1) because this variable is important in explaining smallholders' participation in commercialization (Ochieng et al., 2020; Schulte et al., 2022). Besides, we follow Kim et al. (2019) to calculate the share of days with extreme precipitation (at or higher than 30 mm per day) in a year using the rainfall data from the TRMM. Due to the availability of the TRMM data, we use the two-year lagged share of days with extreme precipitation as the additional IV for $\mathbf{C}_{iw}$. The rationale behind using the lagged share of heavy rain days as the IV of crop commercialization is that the number of days with extreme rainfall reflecting weather shocks can drive the household decision to sell more output to have more cash and to cope with shocks in the current year (Nguyen et al., 2020a).

To construct the IV for $\mathbf{M}_{iw}$ (as $\mathbf{Z}_{iw}$ in Equation (2)), we use the rainfall data from the TRMM. In this regard, we apply the method proposed by Jones and Hulme (1996) to calculate the Standardized Rainfall Anomaly Index (SRAI) for each village in each year relied on the long-term average and standard deviation of annual precipitation between 1998 and 2014. Then,

the two-year lagged SRAI is used as the additional IV for $\mathbf{M}_{iw}$. The SRAI is used as the IV of crop mechanization because the SRAI which implies more general extreme weather events triggers the development of agricultural mechanization as a farm management decision (Liebenehm et al., 2023).

We run several quality checks to validate the appropriateness of these IVs and to check the use of this method. First, since we use rainfall data for both crop commercialization and mechanization costs, additional estimations (first stage estimations) are run to validate if the $\mathbf{R}_{iw}$ has a significant correlation with the $\mathbf{M}_{iw}$ and $\mathbf{Z}_{iw}$ has a significant correlation with the $\mathbf{C}_{iw}$. The results of these additional estimations show that the two-year lagged share of heavy rain days and distance from village to the closest market ($\mathbf{R}_{iw}$) do not have a significant correlation with mechanization costs per ha ($\mathbf{M}_{iw}$) and the two-year lagged SRAI ($\mathbf{Z}_{iw}$) does not have a significant correlation with the ratio of commercialization ($\mathbf{C}_{iw}$) (see Table A2 in the Appendices). Second, we carry out four quality tests for the simultaneous equations model, namely the Hansen-Sargan over-identification statistic, the test of independent equations (Breusch-Pagan Lagrange Multiplier Test), the test of overall system heteroscedasticity (Likelihood Ratio LR Test), and the test of overall system heteroscedasticity (Wald Test). The results of these tests have validated the appropriateness of using this model (see Table A3 in the Appendices). Last, we also check for the problem of multicollinearity among our independent variables using the Variance Inflation Factors (VIF). The results of VIF values show that our model does not suffer from multicollinearity (see Columns 1 and 2 in Table A4 in the Appendices). We follow Bietenbeck et al. (2019) and Laß and Wooden (2023) to use the “*xtdata, fe*” command in STATA to convert the data into a suitable format for FE estimation. We bootstrap our estimates with 1000 replications and cluster them at village level to have robust standard errors.

5.4.2 Estimating the effects of crop commercialization on non-farm employment

In the second step, we investigate the impacts of crop commercialization on smallholders’ non-farm employment. The time aspect of farm production should be taken into account to examine its effect on structural change (Amare et al., 2023; Schulte et al., 2023). Based on the data, we were unable to substantiate whether non-farm employment occurred after crop production. Therefore, we generate a variable that reflects the change in crop commercialization ratio over time. In this context, we use a dummy indicating whether the smallholder experiences an improvement in crop commercialization in this wave (w) compared to the previous wave ($w-1$) (yes = 1; otherwise = 0). In other words, the improvement in crop

commercialization can be calculated as follows:

$$IC_{iw} = 1 \text{ if } \Delta IC_{iw} = C_{iw} - C_{iw-1} > 0 \quad (3)$$

The improvement of crop commercialization (IC_{iw}) is then included in the estimation of structural change to examine the effects of improved crop commercialization. We use a fixed-effects model to account for unobservable (time-invariant) characteristics of smallholders, which can be specified as:

$$SC_{iw} = \theta_0 + \theta_1 IC_{iw} + \theta_2 X_{iw} + h_i + \omega_{iw} \quad (4)$$

where SC_{iw} is an indicator of structural change which is the non-farm income per capita of smallholder i at wave w . IC_{iw} is a dummy variable of an improvement in crop commercialization in this wave compared to the previous wave as generated from Equation 3. X_{iw} is the group of household and village variables which remains unchanged as defined in Equations 1 and 2. h_i is the household fixed-effects and ω_{iw} is the unobserved time-varying idiosyncratic errors.

As explained in Equation 1, IC_{iw} is endogenous in Equation (4) because it can be correlated with household and village characteristics. Furthermore, there might be a reversed causality between commercialization and non-farm employment (Woldeyohanes et al., 2017). We address these problems by applying FE estimation with a control function (CF) approach (Wooldridge, 2015). In this CF approach, at least one IV is included in the first-stage model of the factors affecting smallholders' crop commercialization to predict the residuals (Ogotu et al., 2020). Then, these residuals are entailed in the estimation of Equation (3) as an additional regressor. We specify the FE model with the CF approach for smallholders' structural change as follows:

$$SC_{iw} = \varphi_0 + \varphi_1 IC_{iw} + \varphi_2 X_{iw} + \varphi_3 \text{Residual_}IC_{iw} + h_i + \xi_{iw} \quad (5)$$

In Equation (5), SC_{iw} , IC_{iw} , and X_{iw} remain unchanged as defined in Equation (4). $\text{Residual_}IC_{iw}$ is the residuals of the smallholders' crop commercialization generated from the first-stage estimation with an IV. h_i is the household fixed-effects and ξ_{iw} is the unobserved time-varying errors. To construct the IV for the first-stage estimation, we again use rainfall data. This time, we generate the Standardized Rainfall Anomaly Index (SRAI) for each village in each month based on the long-term average and standard deviation of monthly precipitation between 1998 and 2014 (Jones and Hulme, 1996). This SRAI can be considered as normal if it varies from -1.0 to 1.0, those values which exceed this range can be considered as extreme precipitation. Then, we count the number of months within a year with an extreme rainfall and

use the share of months within a year with an extreme rainfall as the IV for crop commercialization. Again, the two-year lagged share of months with extreme SRAI is used. To validate the requirement of exclusion restrictions, we run additional estimations to check if this IV is only correlated with crop commercialization, but not with non-farm income per capita. The results of the additional estimations presented in Table A5 in the Appendices confirm the validation of the exclusion restrictions.

We conduct several robustness checks for the impacts of an improvement in crop commercialization on non-farm employment. First, besides the estimations using FE and CF, we run additional estimations of Equation (3) using FE with the two-stage-least-squares (2SLS) method. In this FE-2SLS estimation, we use the same IV as in FE-CF models to instrument the IC_{iw} in Equation (5). Second, since the outcome of structural change includes increased productivity (Amare et al., 2023; Timmer, 1988; Üngör, 2013), we use the share of non-farm laborers in total household labors as the dependent variable reflecting the structural change in regressing Equation (5) to examine if the effects of crop commercialization are consistent in a different measurement of structural change.

We check the quality of the IVs in the FE-2SLS models by performing two tests, namely under-identifying test (the rk LM statistics from Kleibergen and Paap (2006)) and weak identifying test (the Kleibergen-Paap rk Wald F statistics from Staiger and Stock (1997)). The results of these tests presented in the “post-estimation” section of Table 5.4 have confirmed the appropriateness of using these IVs in our estimations. We also check for a problem of multicollinearity of the independent variables in Equation (3). The VIF results show no evidence of this problem (see Column 3 in Table A4 in the Appendices). We cluster these estimates at village level to have robust standard errors.

5.4.3 Assessing the distributional effects of crop commercialization on smallholders’ farm income and total income

We complete our empirical strategy by examining how the benefits of crop commercialization are distributed across smallholders’ farm income and total income groups. In other words, if the commercialization of crops has a positive effect on the income of smallholders, then the question arises as to which income groups can benefit more from commercialization. To examine the relative distributional effects of crop commercialization on farm income and total income of smallholder farmers, we use an unconditional quantile regression (UQR) model proposed by Firpo et al. (2009). This UQR model can not only estimate the unconditional partial effects of changes in the distribution of independent variables

on the distribution statistics of the outcome variable through a two-stage procedure (Borgen, 2016), but also consider robust and clustered standard errors in the estimation (Baltagi and Ghosh, 2017; Nguyen and Do, 2022; Nguyen et al., 2023; Rios-Avila, 2020). In the first stage, this UQR model calculates the recentered influence function (RIF) as follows:

$$RIF(\mathbf{INC}; q_\tau, F_P) = q_\tau + \frac{\tau - 1\{\mathbf{INC} \leq q_\tau\}}{f_{\mathbf{INC}}(q_\tau)} \quad (6)$$

where q_τ denotes the value of smallholders' income (either daily farm per capita or daily household income per capita), $\mathbf{INC}$, at the quantile τ . F_P is the cumulative distribution function of the outcome indicator $\mathbf{INC}$, and $f_{\mathbf{INC}}(q_\tau)$ is the density of the outcome $\mathbf{INC}$ at q_τ . $1\{\mathbf{INC} \leq q_\tau\}$ reflects an indicator function to determine if the value of $\mathbf{INC}$ is below q_τ . In the second stage, the UQR model estimates the distributional effects of crop commercialization on smallholders' farm income and total income ($\mathbf{INC}$) as:

$$I[RIF(\mathbf{INC}_i; q_\tau)|\mathbf{X}, \mathbf{IC}] = \Psi + \psi \mathbf{IC}_i + \phi \mathbf{X}_i + \varsigma_i \quad (7)$$

where $\mathbf{IC}_i$ is the improvement of crop commercialization of smallholder i and $\mathbf{X}_i$ denotes the group of household and village control variables as defined in Equations 1 and 2. The relative unconditional partial effects of crop commercialization can be generated from the formula: $\psi = \frac{\partial v(F_P)}{\partial \bar{R}_k}$ as suggested by (Rios-Avila, 2020).

To address the endogeneity of smallholders' crop commercialization ($\mathbf{IC}_i$) in regressing Equation (7), we employ the control function approach with the same IV as explained in Equations (4) and (5). To have robust standard errors, we cluster the estimation of UQR models at the village level.

5.5 Results and discussion

5.5.1 Relationship between crop commercialization and mechanization

We begin by answering the first research question of whether smallholder commercialization can affect structural change by explaining the interdependence of crop commercialization and mechanization. The results from the simultaneous equations model (Model (1) in Table 5.3) show that there is a positive and significant relationship between crop commercialization and mechanization costs in crop production. Our result is consistent with the findings of Liu et al. (2020) and Pingali and Rosegrant (1995) that increasing crop commercialization has a positive influence on mechanization. At the same time, mechanization plays an important role in stimulating crop commercialization (Owusu and İşcan, 2021; Schulte et al., 2022). During commercialization, smallholder farmers shift from non-tradable inputs to

purchased inputs such as technology, and these production technologies simultaneously determine their decision to participate in the market (Abraham et al., 2022; Binswanger and Braun, 1991; Woldeyohanes et al., 2017). Further, the positive interrelationship between crop commercialization and mechanization remains consistent when we use the share of mechanization costs in total production costs as the indicator of mechanization development (Model (2) in Table 5.3).

The remaining household variables include age of head, years of schooling of head, household size, and the number of large transport vehicles, which are negatively associated with crop commercialization. Our results on the age and school years of the household head have some similarities with the findings on the age and education of the household head by Owusu and İşcan (2021). The negative correlation of household size may be due to greater opportunities for the smallholder farmer to engage in other employment activities such as non-farm employment (Do et al., 2022). The influence of large transport vehicle ownership on smallholders' participation in commercialization in our study is different from that of Ochieng et al. (2020). The reason for the negative relationship between the number of transport vehicles and crop commercialization can be attributed to the better use of vehicles for non-farm employment, thereby reducing smallholder farmers' participation in farming (Do et al., 2022).

On the other hand, PSO membership, land area, location of land plots, and number of phones are positively correlated with commercialization. The positive influence of PSO membership can be explained by the benefits of collective action such as lower transaction costs and improved bargaining power (Fisher and Qaim, 2012; Schulte et al., 2022). Our result on land area is consistent with the findings of Owusu and İşcan (2021), Schulte et al. (2022), and Woldeyohanes et al. (2017), as larger farmland area allows smallholder farmers to mechanize their production to achieve higher efficiency (Do et al., 2023; Fukai et al., 2019). Furthermore, the positive impact of phone ownership on smallholder commercialization in our study is consistent with the findings of Ochieng et al. (2020), Schulte et al. (2022), and Zanella (2012).

Regarding village variables, we find that better road quality and the share of households with cable internet at home have a positive correlation, while the number of enterprises in the village and the distance from the village to the provincial capital have a negative correlation with smallholder commercialization. These results are in line with previous studies and have important implications for infrastructure development in rural villages, especially in remote villages (Binswanger et al., 1993; Cazzuffi et al., 2020; Ochieng et al., 2020; Owusu and İşcan, 2021).

Table 5.3. Interrelationship between crop commercialization and mechanization costs.

	Model (1)		Model (2)	
	Crop commercialization	Mechanization costs per hectare (ln)	Crop commercialization	Share of mechanization costs
Mechanization costs per hectare (ln)	1.779*** (0.426)			
Share of mechanization costs			0.129*** (0.032)	
Crop commercialization		5.570*** (1.389)		0.430*** (0.116)
2-year lagged SRAI		-0.085** (0.033)		-0.005* (0.003)
2-year lagged share of heavy rain days	0.831** (0.336)		0.693* (0.420)	
Distance to the closest market	0.000 (0.001)		0.000 (0.001)	
Age of head	-0.003** (0.001)	0.024*** (0.007)	-0.005*** (0.002)	0.003*** (0.001)
Male head†	0.027 (0.032)	-0.202 (0.208)	0.047 (0.035)	-0.026 (0.016)
Years of schooling of head	-0.014*** (0.005)	0.099*** (0.026)	-0.016*** (0.005)	0.009*** (0.002)
Average years of schooling of adult members	-0.002 (0.003)	0.017 (0.020)	0.003 (0.003)	-0.002 (0.001)
Household size	-0.013** (0.006)	0.091** (0.038)	-0.011* (0.006)	0.005* (0.003)
Share of laborers	-0.003 (0.035)	0.099 (0.231)	-0.093* (0.048)	0.057*** (0.018)
Member of PSO†	0.032* (0.018)	-0.217* (0.115)	0.042** (0.017)	-0.022*** (0.008)
Ethnic majority†	0.073 (0.060)	-0.506 (0.400)	0.013 (0.069)	-0.004 (0.035)
Buddhist head†	0.004 (0.055)	-0.013 (0.371)	-0.021 (0.038)	0.013 (0.018)
Total land area	0.009*** (0.003)	-0.066*** (0.017)	0.016*** (0.005)	-0.009*** (0.002)
All land plots in the same village†	0.025** (0.012)	-0.135 (0.086)	0.040*** (0.013)	-0.019*** (0.006)
Number of phones	0.004 (0.005)	-0.038 (0.032)	0.014** (0.007)	-0.008*** (0.003)
Number of large transport vehicles	-0.031* (0.018)	0.219* (0.117)	-0.036 (0.023)	0.019* (0.011)
Asset poor household†	0.026 (0.019)	-0.214* (0.116)	-0.009 (0.018)	0.004 (0.009)
Number of enterprises in village	-0.008 (0.009)	0.052 (0.061)	0.002 (0.009)	-0.002 (0.004)
Having paved roads in village†	0.022 (0.028)	-0.109 (0.193)	0.045** (0.020)	-0.021** (0.010)
Share of households with access to electricity in village	-0.116*** (0.033)	0.856*** (0.143)	-0.142*** (0.039)	0.077*** (0.012)
Share of households with phone line at home in village	-0.004 (0.096)	-0.010 (0.639)	0.030 (0.108)	-0.019 (0.054)
Share of households with cable internet at home in village	0.108 (0.092)	-0.634 (0.654)	0.125 (0.081)	-0.058 (0.044)
Distance to provincial capital	-0.001 (0.001)	0.006 (0.004)	-0.001** (0.001)	0.001** (0.000)
Constant	0.326** (0.139)	-1.742 (1.066)	0.561*** (0.153)	-0.268*** (0.086)
Number of observations	8601		8601	
Wald chi2	46.630		52.369	
Prob > chi2	0.002		0.000	

Note: Robust standard errors bootstrapped with 1000 replications and clustered at village level in parentheses; ln: natural logarithm; †: Dummy variable; *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

5.5.2 Effects of crop commercialization on non-farm employment

Table 5.4 summarizes the results of the effects of crop commercialization on non-farm income per capita from fixed-effects with control function (FE-CF) and fixed-effects with instrumental variable (FE-2SLS) estimations. The results of both FE-CF and FE-2SLS estimations suggest that improving crop commercialization has a significant and positive effect on non-farm income per capita. While empirical evidence on the effects of crop commercialization on structural change at the smallholder level is limited, our findings provide important evidence on the role of commercialization in influencing structural change in agriculture and confirm that smallholder-led crop commercialization drives structural change in agriculture at the micro level. As a robustness check, the impact of improving crop commercialization on the share of non-farm laborers remains positive and significant from both FE-CF and FE-2SLS estimations.

Table 5.4. Effects of improving crop commercialization on non-farm employment.

	Non-farm income per capita (ln)			Share of non-farm laborers		
	FE-CF	FE-2SLS		FE-CF	FE-2SLS	
		First stage	Second stage		First stage	Second stage
An improvement in crop commercialization	1.503* (0.911)		1.210* (0.735)	0.747*** (0.068)		0.600*** (0.099)
2-year lagged number of months with extreme rainfall		0.317*** (0.040)			0.317*** (0.040)	
Household control	Yes	Yes	Yes	Yes	Yes	Yes
Village control	Yes	Yes	Yes	Yes	Yes	Yes
Residuals of an improvement in crop commercialization	-1.531 (0.930)			-0.767*** (0.069)		
Constant	-2.215* (1.298)	0.515** (0.225)	-2.270* (1.353)	-0.460*** (0.107)	0.515** (0.225)	-0.487*** (0.185)
Number of observations	5734	5734	5734	5734	5734	5734
F statistics	6.56	6.50		38.46	6.50	
Prob > F	0.000	0.000		0.000	0.000	
Wald chi2(21)			176.14			277.78
Prob > chi2			0.000			0.000
Under-identification			0.000			0.000
Weak identification			61.721			61.721

Note: Robust standard errors clustered at village level in parentheses; †: Dummy variable; *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$; The test of under-identification is a LM test relied on the rk LM statistics. The null hypothesis of this LM test is that the model is under-identified. The presented values of these under-identification tests are p-values. The values of weak identification tests are the Kleibergen-Paap rk Wald F statistics. Full results presented in Table A6 in the Appendices.

5.5.3 Distributional effects of crop commercialization on farm income and total income of smallholders

Table 5.5 presents the distributional effects of crop commercialization on five quantile groups of smallholder farm income per capita and total income per capita from UQR models. The results show that crop commercialization has a positive impact on farm income per capita

and total income per capita across all income quantiles. In absolute terms, the impacts on both farm income per capita and total income per capita have the greatest value at the 90th income quantile. These results are consistent with previous studies that commercialization of crops brings more benefits for the better-off groups (Ogotu and Qaim, 2019; Tabe Ojong et al., 2022).

However, if we consider the relative aspect of the impacts, crop commercialization appears to bring more benefits to smallholder farmers in the poorest groups (10th and 25th income quantiles). In particular, improving crop commercialization can increase the per capita farm income of smallholders in the 10th and 25th quantile groups by about 25% and 5.6%, respectively. For total income, an improvement in crop commercialization leads to an increase in total income per capita by 10.2% for the 10th income quantile and by 5.8% for the 25th income quantile. Hence, crop commercialization can play an important role in increasing income and reducing income inequality.

Table 5.5. Distributional effects of improving crop commercialization on farm and household income of smallholders.

	Income quantiles				
	10 th	25 th	50 th	75 th	90 th
<i>Panel A. Effect on daily farm income per capita</i>					
Effect of improving crop commercialization	1.697*** (0.220)	2.349*** (0.259)	3.943*** (0.457)	6.689*** (1.149)	12.688*** (3.064)
Sample mean RIF	0.068	0.422	1.237	3.093	6.455
Magnitude	24.89%	5.56%	3.19%	2.16%	1.97%
<i>Panel B. Effect on daily household income per capita</i>					
Effect of improving crop commercialization	8.509*** (0.719)	11.900*** (0.807)	15.986*** (1.205)	18.909*** (2.135)	23.353*** (5.251)
Sample mean RIF	0.832	2.055	4.397	8.250	14.605
Magnitude	10.23%	5.79%	3.64%	2.29%	1.60%

Note: Robust standard errors clustered at village level in parentheses; *** $p < 0.01$; Full results are presented in Tables A7 and A8 in the Appendices.

5.6 Summary and policy implications

We examine the influence of crop commercialization on the structural change of smallholder farmers and assess the distributional effects of crop commercialization on smallholder farm and total income based on a balanced panel of 2,867 rural smallholders collected in three survey waves from Thailand and Vietnam. For the first research question, we consider two indicators of structural change, namely mechanization costs and non-farm per capita income, as proxies for the development of mechanization and the shift to non-farm

employment, respectively. We use a simultaneous equations model to examine the simultaneity of crop commercialization and mechanization. The results of the simultaneous equations model show that crop commercialization has a positive and significant relationship with crop mechanization. The results remain consistent if we consider the share of mechanization costs in the total cost of crop production. In addition, we use two econometric methods, namely the fixed-effects with controls and the fixed-effects with instrumental variables to address the endogeneity of an improvement in crop commercialization. The results of these estimations suggest that improved commercialization of crops has a positive and significant effect on non-farm per capita income. Our results are robust if we replace the dependent variable with the share of non-farm laborers in total household laborers. For the second research question, we use an unconditional quantile regression (UQR) model to estimate which quantile groups of smallholder farmers benefit more from crop commercialization in terms of farm income and total income. The results show that an improvement in crop commercialization benefits the poorest smallholders in the 10th and 25th income quantile groups more than the better-off, both in terms of farm income and total income per capita.

Several policy implications arise from our empirical findings. To achieve more sustainable, resilient and inclusive growth, these policy suggestions should come in a bundle rather than as separate options (Barrett et al., 2020). First, the commercialization of smallholders should be facilitated to stimulate structural change. Facilitation may relate to (i) supporting mechanization (Diao et al., 2016; Liu et al., 2020), (ii) supporting smallholder collective organization (Barrett, 2008; Fisher and Qaim, 2012), and (iii) increasing farm size either by supporting land renting or improving land consolidation (Abraham et al., 2022; Do et al., 2023; Schulte et al., 2022). Second, commercialization can also be the target of interventions in development programs and policies to combat poverty and reduce inequality, as the poorest smallholder farmers benefit most from commercialization. A special focus should be placed on poor smallholders in order to increase their market participation (Fischer and Qaim, 2012).

Finally, investments in infrastructure such as road quality (e.g., at least paved roads) should be encouraged at the village level (Owusu and İşcan, 2021; Woldeyohanes et al., 2017). Local governments can place a special focus on those villages that are far from urban cities (Barrett, 2008; Cazzuffi et al., 2020; Fisher and Qaim, 2012). Promoting better investments in rural infrastructure provides smallholders with the opportunity not only to commercialize their crops on a larger scale but also to engage in better income-generating activities, thereby stimulating structural change (Banerjee et al., 2020; Queiroz et al., 2021).

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Appendices

Appendix 5.A1. Measurement and definition of variables at household and village levels.

Variables	Measurement	Definition
<i>A. Household level</i>		
<i>A1. Commercialization, structural change, and income</i>		
Crop commercialization	Percentage (%)	The ratio of sold crop products to the total value of crop production
Mechanization costs	PPP\$ (adjusted to 2005 prices)	Mechanization costs for land preparation and harvest per hectare
Share of mechanization costs	Percentage (%)	The percentage of mechanization costs for land preparation and harvest in total production costs
Non-farm income	PPP\$ (adjusted to 2005 prices)	Income from wage-employment and self-employment per non-farm laborer
Share of non-farm laborers	Percentage (%)	The percentage of household members engaged in non-farm employment in total household labor
Daily farm income per capita	PPP\$ (adjusted to 2005 prices)	Household's per capita daily income from crop and livestock activities
Daily household income per capita	PPP\$ (adjusted to 2005 prices)	Household's per capita daily income from all activities
<i>A2. Human capital</i>		
Age of head	Years of age	Ages of the household head
Gender of head	Dummy	Male-headed household = 1; otherwise = 0
Education of head	Years of schooling	Number of years of schooling of the household head
Education of adult members	Years of schooling	Average years of schooling of adult members in the household
Household size	Number of persons	Number of nucleus members in the household
Share of laborers	Percentage (%)	The share of household members in working ages and having ability to work
<i>A3. Social capital</i>		
Member of political and social organization (PSO)	Dummy	If the household head is a member of a political or social organization = 1; otherwise = 0
Ethnicity of head	Dummy	If the household belongs to Thai majority (Thailand) or Kinh majority (Vietnam) = 1; otherwise = 0
Religion of head	Dummy	If the household head is Buddhist = 1; otherwise = 0
<i>A4. Natural capital</i>		
Total land area	hectares (ha)	Total land area of the household
All land plots in the village	Dummy	If all land plots of the household are located in the same as the current village = 1; otherwise = 0
<i>A5. Physical and financial capital</i>		
Number of phones	Quantity	Number of phones (mobile and smartphones) that the household owns
Number of large transport vehicles	Quantity	Number of pickups and trucks that the household owns
Asset value per capita	PPP\$ (adjusted to 2005 prices)	Total asset value per capita of the household including productive and non-productive assets
Asset poor	Dummy	If the household belongs to the 20% poorest group of asset value per capita = 1; otherwise = 0

<i>B. Village level</i>		
Number of enterprises	Quantity	Number of enterprises with more than 9 employees in the village
Having paved roads instead of dirt roads	Dummy	If the main roads in the village are paved roads (instead of dirt roads) = 1; otherwise = 0
Share of households with access to electricity	Percentage (%)	The percentage of households with electricity from national power line in the village
Share of households with phone line at home	Percentage (%)	The percentage of households with phone line at home in the village
Share of households with cable internet at home	Percentage (%)	The percentage of households with cable internet at home in the village
Distance to provincial capital	Kilometer (km)	The distance from the village to the provincial center
Distance to the closest market	Kilometer (km)	The distance from the village to the closest market

Appendix 5.A2. Validating estimations of instrumental variables in the simultaneous equations model of crop commercialization and mechanization.

	With 2-year lagged SRAI		With 2-year lagged share of days with heavy rain	
	Crop commercialization	Mechanization costs per hectare (ln)	Crop commercialization	Mechanization costs per hectare (ln)
2-year lagged SRAI	-0.006 (0.003)	-0.117*** (0.030)		
2-year lagged share of days with heavy rain			0.526** (0.238)	-2.309 (1.987)
Distance to the closest market			-0.000 (0.003)	0.008 (0.018)
Age of head	0.000 (0.001)	0.026*** (0.006)	0.001 (0.001)	0.028*** (0.007)
Male head†	-0.001 (0.018)	-0.207 (0.191)	-0.000 (0.018)	-0.214 (0.189)
Years of schooling of head	-0.003 (0.002)	0.083*** (0.023)	-0.003 (0.002)	0.082*** (0.023)
Average years of schooling of adult members	-0.000 (0.002)	0.017 (0.017)	0.000 (0.002)	0.017 (0.017)
Household size	-0.006* (0.003)	0.059* (0.032)	-0.006** (0.003)	0.054* (0.032)
Share of laborers	0.049*** (0.019)	0.372** (0.183)	0.048*** (0.018)	0.393** (0.185)
Member of PSO†	0.008 (0.010)	-0.170* (0.094)	0.008 (0.010)	-0.184** (0.093)
Ethnic majority†	0.038 (0.037)	-0.295 (0.360)	0.036 (0.037)	-0.284 (0.358)
Buddhist head†	0.002 (0.024)	0.000 (0.318)	0.002 (0.024)	-0.019 (0.318)
Total land area	0.003*** (0.001)	-0.049*** (0.012)	0.003*** (0.001)	-0.047*** (0.012)
All land plots in the same village†	0.028*** (0.007)	0.021 (0.073)	0.030*** (0.007)	0.040 (0.073)
Number of phones	-0.006** (0.003)	-0.072*** (0.027)	-0.005** (0.003)	-0.068** (0.027)
Number of large transport vehicles	-0.008 (0.010)	0.172* (0.103)	-0.012 (0.010)	0.148 (0.103)
Asset poor household†	-0.005 (0.010)	-0.244** (0.099)	-0.004 (0.010)	-0.238** (0.099)
Number of enterprises in village	-0.003 (0.004)	0.036 (0.052)	-0.002 (0.004)	0.047 (0.052)
Having paved roads in village†	0.034*** (0.013)	0.081 (0.153)	0.041*** (0.012)	0.145 (0.151)
Share of households with access to electricity in village	-0.003 (0.012)	0.837*** (0.116)	-0.012 (0.012)	0.808*** (0.117)
Share of households with phone line at home in village	0.002 (0.046)	-0.000 (0.477)	0.002 (0.046)	0.044 (0.496)
Share of households with cable internet at home in village	0.144*** (0.055)	0.166 (0.429)	0.158*** (0.054)	0.388 (0.433)
Distance to provincial capital	-0.000 (0.000)	0.004 (0.004)	-0.000 (0.000)	0.004 (0.004)
Constant	0.443*** (0.075)	0.726 (0.746)	0.407*** (0.078)	0.563 (0.774)
Number of observations	8601	8601	8601	8601
F statistics	4.42	9.66	4.12	7.80
Prob > F	0.000	0.000	0.000	0.000

Note: Robust standard errors clustered at village level in parentheses; †: Dummy variable; ln: natural logarithm; *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Appendix 5.A3. Quality tests of the simultaneous models of crop commercialization and mechanization.

	chi2	Prob.>chi2
<i>A. The model of crop commercialization and mechanization costs per hectare</i>		
Hansen-Sargan over-identification statistic	0.314	0.957
Tests of independent equations (Breusch-Pagan Lagrange Multiplier Test)	8418.835	0.000
Tests of Overall System Heteroscedasticity (Likelihood Ratio LR Test)	3.32e+04	0.000
Tests of Overall System Heteroscedasticity (Wald Test)	3.45e+04	0.000
<i>B. The model of crop commercialization and share of mechanization costs</i>		
Hansen-Sargan over-identification statistic	2.386	0.496
Tests of independent equations (Breusch-Pagan Lagrange Multiplier Test)	8474.067	0.000
Tests of Overall System Heteroscedasticity (Likelihood Ratio LR Test)	3.63e+04	0.000
Tests of Overall System Heteroscedasticity (Wald Test)	6.62e+07	0.000

Appendix 5.A4. Variance Inflation Factor (VIF) values.

	Simultaneous equations model		Effect of an improvement in crop commercialization
	Crop commercialization equation	Mechanization equation	
	(1)	(2)	
Share of mechanization costs	1.17		
Crop commercialization		1.15	
An improvement in crop commercialization [†]			1.02
2-year lagged SRAI		1.12	
2-year lagged share of heavy rain days	1.10		
Distance to the closest market	1.28		
Age of head	1.19	1.19	1.19
Male head [†]	1.09	1.09	1.08
Years of schooling of head	1.66	1.66	1.60
Average years of schooling of adult members	1.39	1.39	1.36
Household size	1.34	1.34	1.37
Share of laborers	1.23	1.22	1.22
Member of PSO [†]	1.90	1.88	1.89
Ethnic majority [†]	1.36	1.35	1.34
Buddhist head [†]	2.66	2.56	2.46
Total land area	1.29	1.31	1.32
All land plots in the same village [†]	1.05	1.05	1.07
Number of phones	1.18	1.17	1.16
Number of large transport vehicles	1.25	1.26	1.26
Asset poor household [†]	1.18	1.19	1.19
Number of enterprises in village	1.04	1.04	1.04
Having paved roads in village [†]	1.23	1.20	1.17
Share of households with access to electricity in village	1.03	1.03	1.06
Share of households with phone line at home	1.11	1.09	1.16
Share of households with cable internet at home in village	1.13	1.14	1.12
Distance to provincial capital	1.19	1.20	1.20
Mean VIF	1.31	1.30	1.30

Note: †: Dummy variable.

Appendix 5.A5. Validating estimations of instrumental variable of an improvement in crop commercialization.

	An improvement in crop commercialization	Non-farm income per capita (ln)
2-year lagged share of months with extreme rainfall	0.317*** (0.040)	0.383 (0.233)
Age of head	-0.003 (0.002)	0.003 (0.013)
Male head [†]	0.089 (0.056)	0.131 (0.360)
Years of schooling of head	-0.018** (0.007)	0.056 (0.041)
Average years of schooling of adult members	0.002 (0.006)	0.112*** (0.036)
Household size	0.015 (0.011)	0.573*** (0.068)
Share of laborers	-0.097 (0.067)	-0.188 (0.417)
Member of PSO [†]	-0.018 (0.032)	0.039 (0.194)
Ethnic majority [†]	0.186* (0.110)	-0.125 (0.785)
Buddhist head [†]	0.016 (0.074)	0.071 (0.448)
Total land area	0.011** (0.005)	-0.013 (0.029)
All land plots in the same village [†]	0.032 (0.022)	0.050 (0.147)
Number of phones	-0.007 (0.009)	0.011 (0.056)
Number of large transport vehicles	-0.075** (0.033)	0.291 (0.219)
Asset poor household [†]	-0.063* (0.037)	-0.474** (0.236)
Number of enterprises in village	0.019 (0.011)	0.017 (0.077)
Having paved roads in village [†]	0.033 (0.044)	0.535** (0.255)
Share of households with access to electricity in village	0.293*** (0.103)	-0.640 (0.450)
Share of households with phone line at home in village	-0.399*** (0.094)	1.606*** (0.444)
Share of households with cable internet at home in village	0.243* (0.138)	0.319 (0.909)
Distance to provincial capital	-0.002** (0.001)	0.007 (0.005)
Constant	0.515** (0.225)	-1.647 (1.293)
Number of observations	5734	5734
F(21,430)	6.497	6.855
Prob > F	0.000	0.000

Note: Robust standard errors clustered at village level in parentheses; †: Dummy variable; *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Appendix 5.A6. Effects of an improvement in crop commercialization on non-farm income per capita and share of non-farm laborers.

	Non-farm income per capita (ln)			Share of non-farm laborers		
	FE-CF	FE-2SLS		FE-CF	FE-2SLS	
		First stage	Second stage		First stage	Second stage
An improvement in crop commercialization	1.503* (0.911)		1.210* (0.735)	0.747*** (0.068)		0.600*** (0.099)
2-year lagged share of months with extreme rainfall		0.317*** (0.040)			0.317*** (0.040)	
Age of head	0.003 (0.013)	-0.003 (0.002)	0.006 (0.013)	-0.001 (0.001)	-0.003 (0.002)	0.001 (0.002)
Male head†	0.103 (0.357)	0.089 (0.056)	0.024 (0.363)	-0.053** (0.024)	0.089 (0.056)	-0.093** (0.040)
Years of schooling of head	0.060 (0.041)	-0.018** (0.007)	0.077* (0.044)	0.007* (0.004)	-0.018** (0.007)	0.016*** (0.006)
Average years of schooling of adult members	0.109*** (0.036)	0.002 (0.006)	0.109*** (0.036)	-0.008*** (0.003)	0.002 (0.006)	-0.008 (0.005)
Household size	0.575*** (0.068)	0.015 (0.011)	0.555*** (0.070)	0.057*** (0.005)	0.015 (0.011)	0.046*** (0.009)
Share of laborers	-0.213 (0.418)	-0.097 (0.067)	-0.070 (0.423)	0.520*** (0.029)	-0.097 (0.067)	0.591*** (0.048)
Member of PSO†	0.006 (0.193)	-0.018 (0.032)	0.060 (0.200)	-0.027** (0.013)	-0.018 (0.032)	0.001 (0.024)
Ethnic majority†	-0.100 (0.786)	0.186* (0.110)	-0.350 (0.833)	0.003 (0.059)	0.186* (0.110)	-0.122 (0.109)
Buddhist head†	0.150 (0.451)	0.016 (0.074)	0.052 (0.472)	0.080** (0.034)	0.016 (0.074)	0.031 (0.057)
Total land area	-0.030 (0.030)	0.011** (0.005)	-0.027 (0.030)	-0.013*** (0.002)	0.011** (0.005)	-0.011*** (0.003)
All land plots in the same village†	0.028 (0.149)	0.032 (0.022)	0.011 (0.153)	-0.041*** (0.010)	0.032 (0.022)	-0.050*** (0.017)
Number of phones	0.014 (0.056)	-0.007 (0.009)	0.020 (0.058)	0.004 (0.004)	-0.007 (0.009)	0.006 (0.007)
Number of large transport vehicles	0.305 (0.218)	-0.075** (0.033)	0.382* (0.215)	0.012 (0.020)	-0.075** (0.033)	0.051* (0.030)
Asset poor household†	-0.371 (0.240)	-0.063* (0.037)	-0.398 (0.247)	0.040** (0.016)	-0.063* (0.037)	0.026 (0.027)
Number of enterprises in village	0.026 (0.077)	0.019 (0.011)	-0.005 (0.072)	-0.001 (0.004)	0.019 (0.011)	-0.016* (0.009)
Having paved roads in village †	0.383 (0.298)	0.033 (0.044)	0.495* (0.265)	-0.169*** (0.020)	0.033 (0.044)	-0.113*** (0.031)
Share of households with access to electricity in village	-0.722 (0.465)	0.293*** (0.103)	-0.995* (0.532)	-0.083** (0.036)	0.293*** (0.103)	-0.219*** (0.083)
Share of households with phone line at home in village	1.811*** (0.470)	-0.399*** (0.094)	2.088*** (0.563)	0.079 (0.065)	-0.399*** (0.094)	0.218** (0.107)
Share of households with cable internet at home in village	0.501 (0.893)	0.243* (0.138)	0.025 (1.035)	-0.106*** (0.041)	0.243* (0.138)	-0.345*** (0.102)
Distance to provincial capital	0.008 (0.005)	-0.002** (0.001)	0.010* (0.005)	0.000 (0.000)	-0.002** (0.001)	0.001* (0.001)
Residuals of an improvement in crop commercialization	-1.531 (0.930)			-0.767*** (0.069)		
Constant	-2.215* (1.298)	0.515** (0.225)	-2.270* (1.353)	-0.460*** (0.107)	0.515** (0.225)	-0.487*** (0.185)
Number of observations	5734	5734	5734	5734	5734	5734
F statistics	6.56	6.50		38.46	6.50	
Prob > F	0.000	0.000		0.000	0.000	
Wald chi2(21)			176.14			277.78
Prob > chi2			0.000			0.000
Under-identification			0.000			0.000
Weak identification			61.721			61.721

Note: Robust standard errors clustered at village level in parentheses; †: Dummy variable; ln: natural logarithm; *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$; The test of under-identification is a LM test relied on the rk LM statistics. The null hypothesis of this LM test is that the model is under-identified. The presented values of these under-identification tests are p-values. The values of weak identification tests are the Kleibergen-Paap rk Wald F statistics.

Appendix 5.A7. Distributional effects of an improvement in crop commercialization on farm income of smallholders.

	Quantiles of daily farm income per capita				
	10 th	25 th	50 th	75 th	90 th
An improvement in crop commercialization [†]	1.697*** (0.220)	2.349*** (0.259)	3.943*** (0.457)	6.689*** (1.149)	12.688*** (3.064)
Age of head	-0.000 (0.001)	-0.001 (0.002)	-0.004 (0.003)	-0.016*** (0.006)	-0.033** (0.014)
Male head [†]	0.051 (0.037)	0.065 (0.042)	0.210*** (0.072)	0.597*** (0.152)	0.697* (0.384)
Years of schooling of head	0.003 (0.005)	0.005 (0.006)	0.024** (0.011)	0.042* (0.025)	0.061 (0.068)
Average years of schooling of adult members	0.002 (0.006)	0.003 (0.007)	0.006 (0.012)	0.035 (0.030)	0.108 (0.093)
Household size	-0.019* (0.010)	-0.045*** (0.012)	-0.158*** (0.020)	-0.488*** (0.047)	-1.082*** (0.120)
Share of laborers	0.021 (0.066)	0.251*** (0.083)	0.395*** (0.138)	0.838** (0.325)	1.810** (0.876)
Member of PSO [†]	0.007 (0.036)	0.006 (0.043)	-0.101 (0.081)	-0.207 (0.177)	-0.393 (0.499)
Ethnic majority [†]	0.127** (0.052)	0.283*** (0.068)	0.439*** (0.122)	0.732** (0.288)	2.400*** (0.652)
Buddhist head [†]	-0.001 (0.043)	0.116** (0.056)	0.210** (0.099)	0.251 (0.240)	-0.976* (0.563)
Total land area	-0.011* (0.007)	0.011 (0.007)	0.078*** (0.016)	0.332*** (0.049)	1.029*** (0.127)
All land plots in the same village [†]	0.031 (0.029)	-0.012 (0.033)	-0.005 (0.059)	0.116 (0.132)	0.149 (0.339)
Number of phones	0.023** (0.011)	0.034** (0.013)	0.085*** (0.024)	0.098* (0.058)	0.172 (0.150)
Number of large transport vehicles	0.006 (0.028)	-0.030 (0.034)	0.029 (0.061)	0.276 (0.169)	0.840* (0.442)
Asset poor household [†]	-0.010 (0.046)	-0.325*** (0.057)	-0.635*** (0.084)	-0.994*** (0.165)	-0.878** (0.349)
Number of enterprises in village	0.002 (0.011)	0.014 (0.014)	0.039 (0.027)	-0.019 (0.068)	0.017 (0.142)
Having paved roads in village [†]	-0.232*** (0.053)	-0.452*** (0.071)	-0.902*** (0.134)	-2.102*** (0.392)	-4.982*** (1.095)
Share of households with access to electricity in village	-0.118 (0.111)	0.024 (0.157)	0.052 (0.301)	-0.313 (0.667)	0.770 (1.248)
Share of households with phone line at home in village	0.545** (0.240)	0.422*** (0.155)	0.495* (0.294)	1.222** (0.586)	1.823 (1.620)
Share of households with cable internet at home in village	-0.063 (0.145)	-0.173 (0.206)	0.515 (0.408)	2.323** (0.940)	7.239*** (2.461)
Distance to provincial capital	0.001*** (0.001)	0.002*** (0.001)	0.006*** (0.001)	0.013*** (0.003)	0.022*** (0.008)
Residuals of an improvement in crop commercialization	-1.566*** (0.219)	-2.115*** (0.261)	-3.603*** (0.462)	-6.121*** (1.175)	-12.008*** (3.162)
Constant	-1.139*** (0.299)	-1.152*** (0.250)	-1.088** (0.443)	-0.060 (0.995)	0.112 (2.460)
Number of observations	5734	5734	5734	5734	5734
F(22,430)	5.585	21.861	46.536	35.546	17.458
Prob > F	0.000	0.000	0.000	0.000	0.000
R ²	0.029	0.095	0.143	0.144	0.120
Sample mean RIF	0.068	0.422	1.237	3.093	6.455

Note: Robust standard errors clustered at village level in parentheses; †: Dummy variable; *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Appendix 5.A8. Distributional effects of an improvement in crop commercialization on total income of smallholders.

	Quantiles of daily household income per capita				
	10 th	25 th	50 th	75 th	90 th
An improvement in crop commercialization [†]	8.509*** (0.719)	11.900*** (0.807)	15.986*** (1.205)	18.909*** (2.135)	23.353*** (5.251)
Age of head	0.005 (0.004)	-0.001 (0.005)	-0.002 (0.006)	-0.000 (0.011)	0.009 (0.027)
Male head [†]	-0.098 (0.100)	-0.026 (0.116)	-0.132 (0.166)	0.063 (0.284)	0.389 (0.780)
Years of schooling of head	0.032** (0.014)	0.084*** (0.018)	0.142*** (0.026)	0.270*** (0.048)	0.734*** (0.138)
Average years of schooling of adult members	0.014 (0.016)	0.059*** (0.019)	0.138*** (0.030)	0.334*** (0.058)	0.495*** (0.160)
Household size	-0.061** (0.029)	-0.157*** (0.036)	-0.414*** (0.048)	-0.970*** (0.086)	-1.831*** (0.211)
Share of laborers	0.319 (0.207)	0.808*** (0.256)	1.978*** (0.344)	3.767*** (0.626)	6.132*** (1.467)
Member of PSO [†]	-0.326*** (0.119)	-0.225 (0.141)	-0.689*** (0.198)	-0.529 (0.334)	-0.853 (0.865)
Ethnic majority [†]	0.927*** (0.168)	1.281*** (0.199)	1.712*** (0.242)	1.737*** (0.384)	0.988 (0.877)
Buddhist head [†]	0.361*** (0.124)	1.096*** (0.165)	1.040*** (0.242)	1.095*** (0.416)	1.057 (1.000)
Total land area	-0.087*** (0.016)	-0.084*** (0.018)	-0.026 (0.029)	0.145*** (0.053)	0.601*** (0.152)
All land plots in the same village [†]	-0.239*** (0.085)	-0.223** (0.095)	-0.330** (0.142)	-0.648*** (0.243)	-1.240** (0.614)
Number of phones	0.217*** (0.033)	0.308*** (0.040)	0.363*** (0.054)	0.415*** (0.098)	-0.079 (0.274)
Number of large transport vehicles	-0.028 (0.065)	0.149** (0.073)	0.683*** (0.120)	2.280*** (0.333)	6.220*** (1.002)
Asset poor household [†]	-0.121 (0.142)	-0.553*** (0.167)	-1.041*** (0.190)	-0.681** (0.277)	-0.623 (0.588)
Number of enterprises in village	0.035 (0.028)	0.057 (0.044)	0.105** (0.052)	0.107 (0.093)	0.699*** (0.261)
Having paved roads in village [†]	-0.807*** (0.204)	-1.544*** (0.223)	-2.104*** (0.319)	-2.818*** (0.561)	-4.407*** (1.316)
Share of households with access to electricity in village	-0.474 (0.347)	-0.067 (0.441)	-0.018 (0.607)	-0.975 (1.036)	1.046 (2.232)
Share of households with phone line at home in village	2.054*** (0.747)	2.297*** (0.470)	2.285*** (0.742)	1.180 (1.294)	1.530 (2.593)
Share of households with cable internet at home in village	0.978** (0.482)	1.649*** (0.462)	3.137*** (0.799)	7.259*** (1.425)	19.203*** (4.389)
Distance to provincial capital	-0.001 (0.002)	-0.001 (0.002)	0.002 (0.003)	0.009* (0.005)	-0.018 (0.012)
Residuals of an improvement in crop commercialization	-8.303*** (0.717)	-11.642*** (0.810)	-15.682*** (1.212)	-18.497*** (2.187)	-22.670*** (5.406)
Constant	-5.248*** (0.842)	-6.688*** (0.759)	-6.842*** (1.030)	-4.308** (1.837)	-3.006 (4.090)
Number of observations	5734	5734	5734	5734	5734
F(22,430)	11.781	41.312	79.472	55.661	20.600
Prob > F	0.000	0.000	0.000	0.000	0.000
R ²	0.076	0.152	0.192	0.173	0.106
Sample mean RIF	0.832	2.055	4.397	8.250	14.605

Note: Robust standard errors clustered at village level in parentheses; †: Dummy variable; *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

CHAPTER 6: WEATHER SHOCKS, IRRIGATION DEVELOPMENT AND POVERTY: EVIDENCE FROM RURAL NORTHEAST THAILAND AND CENTRAL VIETNAM

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CHAPTER 7: FEMALE RURAL-URBAN MIGRANTS AND ONLINE MARKETPLACES IN EMERGING ECONOMIES: EVIDENCE FROM THAILAND AND VIETNAM

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